

Capital, Intangibles, and Financial Frictions*

Fergal Hanks[†] Joao Monteiro[‡]

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Abstract

How do financial frictions shape firms' investment in intangible capital? We show that financial constraints distort firms' input choices, leading to systematic underinvestment in intangible assets. Exploiting an investment subsidy in Portugal that lowered the cost of both physical and intangible capital while keeping their relative price unchanged, we find that treated firms reduced their capital-to-intangible ratio by 12 percent, with larger effects for small, financially constrained firms. The distribution of treatment effects declines sharply across percentiles - firms with higher initial capital-to-intangible ratios adjust the most - revealing the signature of a binding financial wedge. Going beyond average effects, we recover the entire cross-sectional distribution of wedges between the marginal rate of technical substitution and the price ratio. The recovered distribution corresponds to the least distorted economy consistent with the data, showing that only a small share of firms are unconstrained while roughly one-quarter face wedges exceeding twenty percent - providing a direct quantitative map of financial distortions in production.

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[†]University of Cambridge. Email: fergaldeclanhanks@gmail.com.

[‡]EIEF. Email: joao.monteiro@eief.it.

1 Introduction

Intangible capital has become an increasingly central component of firms' production structure. In the United States, the ratio of intangible to physical capital rose from 0.75 in 1975 to nearly 1 by 2021 (Crouzet et al., 2022). This rise in intangible intensity has been closely associated with higher market concentration (Crouzet and Eberly, 2019) and greater market power (De Ridder, 2024), yet the adoption of intangible capital remains uneven across the firm size distribution. Unlike tangible assets, intangible capital is rarely usable as collateral. Its limited excludability makes it difficult to enforce property rights over the surplus it generates, limiting its ability to support debt finance. Consequently, intangible investment is typically financed with internal cash rather than borrowing (Falato et al., 2022; Sun and Xiaolan, 2019). This financing asymmetry implies that firms with limited liquidity systematically underinvest in intangibles. When the returns to intangible capital exceed those of physical capital, financial frictions can therefore lead smaller firms to underinvest, widening productivity and profitability gaps and reinforcing the dominance of larger firms.

This paper examines how financial frictions shape firms' investment in intangible capital. We study the introduction of an investment tax subsidy in Portugal in 2014 that allowed firms to deduct a share of their investment expenditures—whether in physical or intangible assets—from their tax liability. The policy reduced the cost of both inputs while leaving their relative price unchanged, thereby relaxing financial constraints through an increase in internal liquidity. The program's design introduced two sources of quasi-experimental variation. First, eligibility was restricted to specific sectors—such as manufacturing—while excluding others, including most services, agriculture, and mining. Second, the generosity of the subsidy varied across regions: firms in some areas could deduct up to 25 percent of investment costs, whereas those in others were limited to 10 percent.

To frame our empirical approach, we begin with the firm's decision problem in allocating resources between physical and intangible capital. In a frictionless environment, the optimal capital mix equates the marginal rate of technical substitution to the relative price of inputs. Because the tax subsidy reduced the cost of both forms of capital proportionally, it should not alter this ratio. However, in the presence of financial frictions, a wedge emerges between the marginal rate of technical substitution and the relative price of capital. This wedge captures distortions that skew firms' input choices toward physical capital, making the policy an ideal setting to study how liquidity shocks relax such distortions.

We trace the effects of the subsidy using administrative data on Portuguese firms from 2010 to 2019, which provide comprehensive information on balance sheets, sectoral classification, and investment in both physical and intangible capital. To estimate the causal impact of the subsidy, we employ a difference-in-differences framework comparing the evolution of the physical-to-intangible capital ratio among treated and control firms. Between 2014 and 2019, treated firms reduced this ratio by 12 percent relative to control firms.

This decline is inconsistent with the predictions of the frictionless model but aligns closely with a framework featuring financial frictions. In the presence of such frictions, a wedge arises between the marginal rate of technical substitution and the relative price of inputs. By increasing firms' cash holdings, the investment subsidy relaxes these constraints and reduces the wedge, thereby shifting the capital mix even without altering relative prices. Hence, the observed adjustment in firms' capital composition provides direct evidence that financial frictions shape investment decisions. Mapping these findings into our theoretical framework, we infer that the subsidy reduced the wedge between the marginal rate of technical substitution and the price ratio by approximately 12 percent, alleviating capital misallocation. This implies that, prior to the policy, the average firm's capital-to-intangible ratio was at least 12 percent above its frictionless benchmark.

A key identifying assumption in our empirical design is that treated and control firms face similar growth rates in the relative price of physical to intangible capital. If price dynamics differ systematically across groups, our estimates could be biased. To assess this assumption, we exploit regional heterogeneity in the generosity of the subsidy. Because capital markets are national rather than regional, firms across regions should experience similar price trends. Consistent with this, treated and control firms in regions with smaller subsidies exhibit similar post-2014 investment patterns. By contrast, in regions where the subsidy was more generous, treated firms reduced their physical-to-intangible capital ratio by 17 percent relative to control firms. This cross-regional variation strengthens our main result and suggests that the effects are strongest where the policy provided the largest liquidity boost.

Investment responses also vary along the firm size distribution. We find that the decline in the physical-to-intangible capital ratio is driven primarily by small firms, defined as those below the median of the sales distribution. Among small firms, treated firms reduced their ratio by 16 percent relative to control firms, whereas the decline among large firms was only 7 percent. This pattern is consistent with the view that financial frictions bind more tightly for smaller firms, constraining their ability to invest in intangible capital absent policy intervention.

Beyond firms' capital mix, we also examine the extensive margin of adjustment—whether firms change their stock of physical and intangible capital. We find no significant effect on the likelihood of adjusting physical capital, but treated firms are 3 percent more likely than control firms to adjust their stock of intangible capital. Unlike in models with fixed costs of adjustment (Caballero et al., 1995; Khan and Thomas, 2008), the extensive margin in our setting is not driven by non-convex adjustment costs but by the fact that some firms are already at their unconstrained optimum.

The subsidy also affected firms' financial structure and performance. Treated firms reduced short-term debt by 7 percent and increased long-term debt by 6 percent relative to control firms, indicating a shift toward longer-term financing and a more forward-looking balance-sheet structure. Firm performance improved as well: operating profits rose by 1.4 percent relative to control firms, underscoring the broader economic implications of relaxing financial constraints on investment decisions.

We then turn to the distributional implications of the investment subsidy and the subsequent relaxation of financial constraints. To do so, we rely on the *Change-in-Changes* (CiC) estimator of Athey and Imbens (2006) and Chernozhukov, Fernández-Val and Melly (2013), which extends the logic of difference-in-differences from means to entire distributions. The estimator relies on a monotone rank-preservation assumption: absent treatment, the rank of each treated firm within its outcome distribution would evolve through the same monotone transformation observed among control firms. In our context, this assumption holds if the distribution of shocks to relative prices is identical for treated and control firms, as would be the case in standard models with nationally integrated input markets.

Using the CiC framework, we estimate quantile treatment effects for each percentile of the initial distribution of the capital-to-intangible ratio among treated firms. The results are consistent with the presence of financial frictions. Firms in the lower percentiles — those already close to the frictionless input mix — show no statistically significant response following the introduction of the subsidy. In contrast, firms in the upper part of the distribution, which are further from the unconstrained input mix and therefore more likely to be financially constrained, exhibit a substantial decline in the ratio. The downward-sloping pattern of the quantile treatment effects implies a compression in the distribution of wedges, consistent with an increase in allocative efficiency as the most distorted firms move closer to their unconstrained capital mix.

Building on this evidence, we develop a method to recover the cross-sectional distribution of wedges implied by the firm-level responses. Specifically, we use the estimated quantile treatment effects to compute a lower bound for the wedge between the

marginal rate of technical substitution and the price ratio at each percentile, following an approach analogous to [Bau and Matray \(2023\)](#). These lower bounds are then used to construct the distribution of wedges in the economy. This distribution represents the *least distorted economy* consistent with both our theoretical framework and the observed firm-level responses to the investment subsidy. It provides the tightest possible lower bound on misallocation consistent with the data.

The recovered distribution features a discrete mass point at one, corresponding to unconstrained firms. Approximately five percent of firms are located at this point, implying that at most five percent of firms operate without financial frictions. The median wedge is 1.13, almost identical to the lower bound for the average wedge inferred from the event study (1.12). Moreover, distortions are quantitatively significant: twenty-five percent of firms exhibit wedges of at least 1.2, meaning that their capital-to-intangible ratios are at least twenty percent above the frictionless benchmark.

Our recovered distribution is not the only one consistent with the data and the model. Any distribution that first-order stochastically dominates our recovered distribution would also satisfy the empirical and theoretical constraints. However, the distribution we estimate represents the allocation with the *least amount of distortions*—a natural lower bound on misallocation. In doing so, we provide a new way to measure the magnitude and shape of financial distortions in the economy.

Related Literature Our findings contribute to several strands of research on financial frictions, fiscal policy, and intangible investment.

A large literature studies how financial frictions distort resource allocation across firms and reduce aggregate productivity. Seminal contributions include [Hsieh and Klenow \(2009\)](#), [Buera, Kaboski and Shin \(2011\)](#), [Midrigan and Xu \(2014\)](#), [Moll \(2014\)](#), [Greenwood et al. \(2010\)](#), [Bau and Matray \(2023\)](#), and [Ottonello and Winberry \(2020\)](#). In macroeconomic theory, models such as [Kiyotaki and Moore \(1997\)](#) emphasize how credit constraints can amplify and propagate shocks through firms' investment decisions. We build on this insight by showing that financial frictions not only misallocate resources across firms but also distort *within-firm* investment decisions—causing firms to overinvest in physical capital relative to intangible capital, even when the latter offers higher returns. Our analysis highlights that misallocation operates both across firms and across asset types within firms.

Our paper is also related to work examining how fiscal policy affects investment behavior. Following the seminal contributions of [Hall and Jorgenson \(1967\)](#), [Caballero, Engel and Haltiwanger \(1995\)](#), and [Cummins, Hassett and Hubbard \(1994\)](#), a large empirical

literature has shown that tax incentives can stimulate investment and output. [Zwick and Mahon \(2017\)](#) exploit variation in bonus depreciation and find that firms increase investment in response, with effects concentrated among smaller and financially constrained firms. Conversely, [Yagan \(2015\)](#) finds little evidence that dividend tax cuts affect investment, highlighting the importance of policy design. Our setting is closer to an accelerated depreciation scheme: the subsidy reduces the user cost of both tangible and intangible capital. However, because firms choose between two types of capital, the policy also affects the *composition* of investment, not just its level.

A third related literature studies the rise of intangible investment and its financing constraints. [Gutiérrez and Philippon \(2017\)](#) document a decline in aggregate investment, while [Crouzet and Eberly \(2023\)](#) show that this pattern largely reflects the growing importance of intangible assets. As [Crouzet et al. \(2022\)](#) and [Eisfeldt and Papanikolaou \(2013\)](#) emphasize, intangible capital differs from physical capital because it is difficult to collateralize and finance externally. Empirically, [Falato et al. \(2022\)](#) and [Sun and Xiaolan \(2019\)](#) show that intangible investment is predominantly funded with internal cash flows. We contribute to this literature by providing causal evidence that financial frictions create a wedge that leads firms to underinvest in intangible capital, and by showing how a liquidity shock—such as a tax subsidy—can relax this constraint.

The paper most closely related to ours is [Ottonello and Winberry \(2023\)](#), who study how financial constraints shape firms’ trade-off between innovation and expansion. While their focus is on the aggregate implications of this trade-off, we examine smaller firms that invest in intangibles primarily as a form of *technology adoption*. Our findings highlight a complementary mechanism: rather than reallocating effort between innovation and scale, firms adjust the composition of their capital stock by substituting toward intangible assets when financing constraints are relaxed.

Outline. Section 2 explains the shock and describes the data. Section 3 describes the empirical strategy. Section 4 explains how we recover the distribution of wedges. Section 5 concludes.

2 The Portuguese Investment Subsidy

2.1 Description of the Policy Shock

In October 2014, Portugal introduced the *Regime Fiscal de Apoio ao Investimento*, a policy aimed at stimulating corporate investment and employment.¹ The program allows firms to deduct part of their investment from taxable income. Eligible expenditures include both tangible fixed assets and intangible capital.

The subsidy was not universal. Eligibility was restricted to a limited set of sectors, including mining, manufacturing, hospitality and accommodation, and technical services such as IT and R&D.² Firms outside these sectors form the control group, as they were not eligible for the subsidy.

The generosity of the subsidy varied across regions. Firms investing in Lisbon or the Algarve could deduct up to 10 percent of investment costs, while firms elsewhere could deduct up to 25 percent. In the analysis, we proxy investment location by the firm's registered address.³ The sectoral eligibility cutoff allows comparisons between treated and untreated firms before and after the policy, while the regional variation in generosity provides an additional source of heterogeneity that we exploit in the analysis.

Firms are not required to claim the deduction in the year the investment occurs. If a firm reports negative taxable income or if its tax liability is smaller than the available deduction, the unused amount can be carried forward for up to two years. This provision is relevant, as roughly one third of Portuguese firms report negative net income.

The policy lowers the cost of both tangible and intangible capital without affecting their relative price, since it applies equally to both forms of investment. It also generates a direct shock to firms' net worth: holding investment policy fixed, end-of-period net worth increases mechanically. This mechanism is illustrated in Table I.

Table I considers a firm with constant sales and operating costs and reports free cash flow at the end of the period—that is, the cash generated by operations. In year 0, the firm invests 20 in either physical or intangible capital, fully financing the expenditure upfront. This investment increases depreciation in subsequent years.⁴

¹Law-Decree 162/2014, dated October 31, established the subsidy. We treat 2015 as the first full year of implementation. The regime combined a corporate income tax deduction with exemptions from property taxes (for up to ten years), the real estate transfer tax, and stamp duty.

²Online Appendix Table A.I lists all eligible sectors. Online Appendix Figure A.1 reports the share of treated firms across employment, wages, sales, capital stock, and firm count. Treated firms account for roughly one quarter of all firms but represent about one third of total wages, sales, and capital stock.

³Most Portuguese firms operate a single establishment. Among multi-establishment firms, the majority concentrate operations within a single region.

⁴Depreciation schedules follow standard tax rules and are unaffected by the subsidy.

Without the subsidy, the only tax benefit from investing arises from future depreciation, which gradually reduces taxable income. With the subsidy, however, the firm can immediately deduct 5 from taxable income in year 0, increasing available cash by the same amount. The resulting rise in cash flow raises firm value one-for-one. As a result, both available cash and net worth increase in response to the subsidy.

In sum, the *Regime Fiscal de Apoio ao Investimento* creates a policy shock that is neutral in relative prices but expansionary in liquidity, effectively relaxing firms' financing constraints. Regional variation in generosity further provides cross-sectional heterogeneity that sharpens identification in the empirical analysis.

2.2 Data Sources

Our primary data source is the *Informação Empresarial Simplificada* (IES), a comprehensive administrative dataset covering the universe of Portuguese firms from 2010 to 2019. The IES is a joint initiative of the Ministry of Justice, Statistics Portugal, and the Bank of Portugal. The data are subject to formal consistency checks by the tax authorities and Statistics Portugal, ensuring comparability across firms and years. All incorporated firms are legally required to file standardized balance sheet and income statement information annually, providing near-complete coverage of the formal corporate sector.

The dataset provides detailed firm-level information, including industry classification, geographic location, and disaggregated measures of both the stock and flow of investment in tangible and intangible capital. These data allow us to construct consistent measures of physical and intangible investment, cash flow, leverage, and profitability, which are central to the empirical analysis that follows. We exclude sole proprietorships, which are not required to report PP&E and intangible assets separately. We also drop firms with non-positive values for sales, total assets, intangible assets, or PP&E.

Table II reports summary statistics for firms in both treated and control sectors. The dataset includes approximately 300,000 firms per year, yielding 2.7 million firm-year observations. The average firm is observed for 5.5 years, reflecting moderate attrition in the panel.⁵

In 2014, the year before the introduction of the investment subsidy, 24 percent of firms operated in treated sectors. Treated firms tend to be larger: their average sales are 38 percent higher than those of control firms, and they employ roughly twice as many workers. By contrast, leverage, productivity (measured as EBITDA-to-sales), and cash holdings do

⁵Most of the analysis relies on this unbalanced panel. Restricting the sample to a balanced panel yields similar results.

not differ significantly between groups. Treated firms also hold more physical capital but report a smaller stock of intangible assets relative to control firms.

Our goal is to understand how the investment subsidy affects the distribution of the capital-to-intangible ratio. Figure 1 plots the distribution of the logarithm of this ratio for all firms in 2014. On average, treated firms hold more physical capital relative to intangible assets, and most of the difference arises from the left tail of the distribution. The variance of the logarithm of the capital-to-intangible ratio is large for both groups: for treated firms, it equals 92 percent of the cross-sectional mean, and for control firms, 93 percent.

To assess how these differences may interact with wedges, we first document how much of the observed heterogeneity arises within versus across sectors. The large variance in the capital-to-intangible ratio documented in Figure 1 motivates a simple decomposition of this variation. Although we formally introduce the theoretical structure in the next section, here we provide an empirical preview of the underlying heterogeneity. The variance may reflect differences in production structure, relative prices, or wedges between the marginal rate of technical substitution and relative prices. Using data through 2014, we regress the logarithm of the capital-to-intangible ratio on a series of fixed effects and report the resulting R^2 values in Table III.

Column (1) shows that differences across sectors—defined at the four-digit industry level—explain only 16 percent of the variation in the capital-to-intangible ratio. Allowing for time-varying sectoral effects, as in column (2), does not raise the explanatory power. If cross-sector differences in production functions were the main source of variation, we would expect a larger share of the variance to be explained given our narrow industry classification. In contrast, firm-level differences within sectors, shown in column (3), account for most of the variation in the capital-to-intangible ratio. This pattern suggests—without yet imposing a structural interpretation—that differences in relative prices or financial wedges are the primary sources of heterogeneity in the data.

2.3 Aggregate Effects

We begin by examining aggregate patterns in investment composition before turning to firm-level outcomes. The analysis focuses on the effect of the investment subsidy on the ratio of physical to intangible capital. Figure 2 plots the aggregate stocks of physical and intangible capital for treated and control firms over time. The series move in parallel before the introduction of the subsidy and diverge thereafter, showing a clear break in trend following the policy.

The subsidy led to increases in both physical and intangible capital at the aggregate level, with treated firms experiencing larger gains than control firms. The expansion in intangible capital, however, was substantially stronger. Between 2014 and 2019, treated firms increased their stock of physical capital by 16 percent more than control firms, while the corresponding increase in intangible capital was 38 percent.⁶ As a result, the ratio of physical to intangible capital declined by 22 percent, as shown in Online Appendix Figure A.3.

This decline is difficult to reconcile with a frictionless model of the firm, as the relative prices of physical and intangible capital remained unchanged. Instead, the shift in the input mix is consistent with the presence of financial frictions, which create a wedge between the marginal rate of technical substitution and relative input costs.

The observed changes reflect not only higher capital stocks but also greater investment activity. Online Appendix Figure A.4 shows that, relative to control firms, treated firms increased investment in physical capital by 4 percentage points and in intangible capital by 18 percentage points. Moreover, treated firms became more likely to invest in intangible capital at all: the probability of any intangible investment rose by 3 percentage points, as shown in Online Appendix Figure A.5.

Taken together, these aggregate patterns indicate that the subsidy expanded overall investment while shifting its composition toward intangibles—a reallocation consistent with a liquidity-driven response rather than a change in relative prices.

3 The Effect on Firm-Level Investment

This section brings the theory to the data. We first lay out a minimal framework in which a price-neutral, liquidity-expanding subsidy affects the capital mix only through a wedge between the marginal rate of technical substitution and relative input prices. Guided by this structure, we estimate an event-study that traces the dynamics of the capital-to-intangible ratio. Next, we turn from composition to levels to assess whether the reallocation reflects substitution or expansion. We conclude with reduced-form evidence on firm performance, reporting how sales, employment, profitability.

⁶The increase in intangible capital is not driven by goodwill. Online Appendix Figure A.6 shows that it is largely accounted for by software acquisitions and patent production.

3.1 A Simple Model

Before estimating the effects of the investment subsidy on the firm-level capital-to-intangible ratio, it is useful to outline a simple theoretical structure to guide the empirical analysis. Consider a firm that combines physical capital k and intangible capital h according to a production function $zf(k, h)$, where z denotes firm-specific productivity and $f(\cdot)$ is concave.⁷ The firm is a price taker in all markets. The elasticity of substitution between physical and intangible capital, $\sigma > 0$, is defined as

$$\sigma \equiv -\frac{d \log(k/h)}{d \log(f_k(k, h)/f_h(k, h))}.$$

Without loss of generality, firm optimality implies

$$\frac{f_k(k, h)}{f_h(k, h)} = \frac{p_k}{p_h} \cdot \frac{1}{\omega},$$

where the left-hand side denotes the marginal rate of technical substitution and the right-hand side equals the relative price of inputs scaled by a wedge $\omega \geq 1$. The wedge captures distortions between the marginal rate of substitution and relative prices. For unconstrained firms, $\omega = 1$; for constrained firms, $\omega > 1$, implying a capital-to-intangible ratio k/h that is too high relative to relative prices. In what follows, we remain agnostic about the source of ω , as in [Hsieh and Klenow \(2009\)](#), and later provide evidence that it reflects financial frictions.⁸

Unlike the standard approach in the misallocation literature, which infers wedges from cross-firm differences in marginal products and therefore requires assumptions about identical technologies, productivity levels, and output prices, our formulation focuses on the marginal rate of technical substitution within the firm. This focus allows us to identify distortions in relative input use without imposing restrictions on the comparability of production technologies or price levels across firms.

If the production function takes the Cobb–Douglas form $f(k, h) = zk^{\alpha}h^{(1-\alpha)\nu}$, where $\alpha \in [0, 1]$ and $\nu \leq 1$, the firm's first-order condition can be written as

$$\frac{k}{h} = \frac{\alpha}{1-\alpha} \cdot \frac{p_h}{p_k} \cdot \omega.$$

⁷We assume $f_k(\cdot), f_h(\cdot) > 0$, $f_{kk}(\cdot), f_{hh}(\cdot) < 0$, and $f_{kh}(\cdot) > 0$. For simplicity, we also assume the production function is homothetic.

⁸Appendix C provides a simple microfoundation based on collateral constraints that limit the pledgability of intangible assets relative to physical capital.

Differences in the capital-to-intangible ratio across firms therefore arise from differences in the capital share α , in relative prices, or in the wedge ω .

Using the firm’s first-order condition and the definition of the elasticity of substitution, the change in the capital-to-intangible ratio can be written as

$$d \log \frac{k}{h} = -\sigma d \log \frac{p_k}{p_h} + \sigma d \log \omega. \quad (1)$$

To a first-order approximation, the firm’s capital-to-intangible ratio changes only in response to movements in relative prices or in the wedge.

We now turn to the investment subsidy. Some firms are exposed to the policy, while others are not. The subsidy does not directly affect the relative price of physical and intangible capital, although general equilibrium effects could alter prices. We assume that any changes in relative prices are identical for treated and control firms.⁹ The subsidy also relaxes firms’ financing constraints by increasing cash holdings. If the elasticity of substitution is identical across firms, changes in the wedge can therefore be identified from the average treatment effect on the capital-to-intangible ratio.¹⁰

The structure also delivers a direct test of the mechanism—the presence of the wedge. We assume that the wedge reflects financial frictions. In particular, following [Ottonello and Winberry \(2023\)](#), we posit that firms can finance physical investment with debt but must use internal funds to finance intangible investment. This financing asymmetry raises the effective cost of intangible capital and tilts the input mix toward physical capital. Consequently, we should observe larger adjustments in the capital-to-intangible ratio among firms that are more financially constrained.

In sum, this framework clarifies how a liquidity shock that leaves relative prices unchanged can alter firms’ input mix through financial wedges. The investment subsidy effectively relaxes these wedges, leading constrained firms to reallocate toward intangible capital. In the next subsection, we bring this framework to the data and estimate the dynamics of this adjustment using an event–study design.

⁹One way to motivate this assumption is to assume that markets for physical and intangible capital are integrated and that firms do not have heterogeneous market power in these markets.

¹⁰Our framework also accommodates heterogeneity in the elasticity of substitution. In that case, and assuming the change in relative prices is the same for treated and control firms, the average treatment effect is given by

$$\mathbb{E}[\sigma] \times \mathbb{E}[d \log \omega] + \text{Cov}(\sigma, d \log \omega).$$

If the elasticity of substitution is not systematically correlated with the change in the wedge induced by the subsidy, we recover the product of the average elasticity and the average change in the wedge. Otherwise, the estimates are biased.

3.2 Effects on Capital-to-Intangible Ratio

We estimate the average treatment effect of the investment subsidy on the ratio of physical to intangible capital using the following event-study specification:

$$\log \left(\frac{k_{i,t+1}}{a_{i,t+1}} \right) = \alpha_i + \lambda_t + \beta X_{i,t} + \sum_{m=-5, m \neq -1}^4 \delta_m \mathbf{1}\{i \in \text{Treated}\} \mathbf{1}\{t = 2015 + m\} + \varepsilon_{i,t}, \quad (2)$$

where the dependent variable is the logarithm of the ratio of physical capital $k_{i,t+1}$ to intangible capital $a_{i,t+1}$ for firm i at the end of year t . The specification includes firm fixed effects α_i , year fixed effects λ_t , and a vector of time-varying controls $X_{i,t}$.

The control vector $X_{i,t}$ includes the logarithm of sales, total assets, and employment; leverage; the ratio of cash holdings to total assets; the EBITDA-to-sales ratio; the share of physical capital in total assets; and the logarithm of total sectoral sales.

The coefficients of interest are the event-time indicators δ_m , which measure the differential change in the capital-to-intangible ratio of treated firms relative to control firms, using 2014 ($m = -1$) as the omitted reference year. Standard errors are clustered at the firm level.

Identification of the coefficients δ_m follows directly from the structure outlined above. Year fixed effects absorb aggregate movements in the relative price of physical and intangible capital, ensuring that the estimated δ_m coefficients capture changes in the wedge ω rather than shifts in p_k/p_h in Equation (1). Firm fixed effects control for time-invariant heterogeneity in production technology, preferences, and financing capacity, while time-varying controls account for shocks to productivity, demand, or firm-specific liquidity. In particular, we include controls for sales, leverage, and cash holdings, which help absorb exposure to heterogeneous financial shocks.

Our identification strategy assumes that, conditional on these observables, firms not eligible for the subsidy provide a valid counterfactual for treated firms. In practice, this means comparing treated firms in sectors such as manufacturing with control firms in sectors such as agriculture. While the assumption cannot be tested directly, it implies a falsifiable prediction: treated and control firms should display parallel pre-treatment trends in the capital-to-intangible ratio. We verify this condition in the event-study results below.

The results from estimating equation (2) are shown in Figure 3. We find no statistically significant differences in the capital-to-intangible ratio between treated and control

firms before the introduction of the policy, supporting the validity of the parallel-trends assumption. Following the implementation of the investment subsidy, treated firms reduce their ratio of physical to intangible capital relative to control firms. Between 2014 and 2019, the average treated firm experiences an 11 percent decline in this ratio.

Under the assumption that relative input prices are unaffected by treatment status, this decline reflects a reduction in the distortion wedge ω . In a Cobb–Douglas setting (where $\sigma = 1$), this corresponds directly to an 11 percent decrease in the wedge. The results therefore indicate that the investment subsidy relaxed financing constraints, consistent with the view that financial frictions distort firms’ capital allocations.

The findings are robust to alternative explanations. Online Appendix Figure B.2 shows that the estimated effect is similar across firms with and without taxable profits, consistent with the fact that even unprofitable firms benefit from the subsidy as they may deduct the investment in future taxes. Online Appendix Figure B.1 confirms that the results also hold in a balanced panel, indicating that entry and exit dynamics are not driving the observed reallocation.

Following Bau and Matray (2023), the fact that the wedge is bounded below by one allows us to reinterpret the event–study estimate of the change in the capital–to–intangible ratio as a lower bound on the initial average wedge in the economy. Under a Cobb–Douglas production function, the change in the average capital–to–intangible ratio is identical to the change in the average wedge.¹¹ Because $\omega_i \geq 1$ for each firm, the initial wedge $\omega_{i,0}$ is bounded from below by e^{-d_i} , where d_i denotes the individual treatment effect in logs. Taking expectations, applying Jensen’s inequality, and using the consistency of our average treatment effect estimates yields $\mathbb{E}[\omega_0] \geq e^{-d}$, where d is the estimated average treatment effect.

The estimated decline of $d = -0.11$ in the average capital–to–intangible ratio therefore implies that the average wedge in the economy prior to the subsidy was at least 1.12. In other words, the effective relative cost of intangible capital was at least 12 percent above its frictionless level. Within our framework, this also implies that the capital–to–intangible ratio itself was at least 12 percent above the frictionless benchmark — consistent with a sizable compositional distortion that the policy subsequently alleviated.

Taken together, these results show that the investment subsidy substantially reduced distortions in firms’ input choices, narrowing the wedge that constrained intangible investment. In the next section, we examine how this adjustment varies across firms with different financial positions and balance-sheet characteristics.

¹¹This equivalence holds even if capital shares are heterogeneous across firms, provided that these shares do not change systematically with treatment over time.

3.3 Heterogeneous Effects

We next examine how the effects of the investment subsidy vary across regions and firms. Heterogeneity in the treatment response is informative along two dimensions. First, regional variation in the generosity of the subsidy allows us to test whether our results could be driven by changes in relative prices. Second, variation across firms provides a direct test of the mechanism: if the wedge ω reflects financial frictions, we should observe stronger effects among financially constrained firms.

We begin by exploiting the policy's regional variation in generosity. Firms located in Lisbon and the Algarve could deduct up to 10 percent of investment costs, whereas firms elsewhere could deduct up to 25 percent. We re-estimate equation (2) separately for high- and low-subsidy regions, and present the results in Figure 4.

Among firms in low-subsidy regions, we find no statistically significant change in capital mix. In contrast, treated firms in high-subsidy regions experience a 17 percent decline in the capital-to-intangible ratio relative to control firms between 2014 and 2019. This pattern suggests that small subsidies are insufficient to induce firms to adjust their capital composition, while larger subsidies meaningfully relax financial constraints. The result is consistent with models featuring fixed adjustment costs—such as Khan and Thomas (2008)—in which small shocks fail to overcome inaction thresholds.

Regional heterogeneity also helps rule out alternative explanations based on differential price movements. If the relative price of intangible capital had fallen more sharply for treated firms—perhaps because of sector-specific input demands—we would expect to see similar effects in all regions. Instead, we observe large effects only in high-subsidy areas. Given that capital markets are nationally integrated, this pattern is difficult to reconcile with price-based explanations and instead supports the interpretation that the subsidy operated through liquidity rather than prices.

We then examine heterogeneity across firms. The theoretical framework predicts that the investment subsidy should be most effective for financially constrained firms, as these firms face higher wedges between the marginal rate of substitution and relative prices. To test this prediction, we split the sample by firm size—measured by sales in 2014—as a proxy for financing constraints. Firms below the median are classified as small (more likely constrained), while firms above the median are classified as large (less likely constrained). Figure 5 reports the results.

Small treated firms experience a 16 percent decline in the capital-to-intangible ratio relative to small control firms between 2014 and 2018.¹² In contrast, large treated firms

¹²These results are robust to alternative definitions of firm size. Online Appendix Figure B.3 shows that similar conclusions hold when using total assets or physical capital stock.

exhibit a more modest 7 percent decline relative to their control counterparts. These patterns reinforce the view that financial frictions distort firms' capital allocation and that the investment subsidy was especially effective at relaxing these distortions among smaller, more constrained firms.

Taken together, these results strengthen our interpretation of the aggregate evidence. The regional analysis rules out explanations based on relative price changes, while the firm-level results confirm that financial constraints drive the observed reallocation. The shift toward intangible capital therefore reflects a relaxation of financing wedges rather than differences in production technology or market prices.

3.4 Effects on the Levels of Physical and Intangible Capital

Thus far, we have focused on how firms adjust the composition of capital—specifically, the trade-off between physical and intangible assets. We now turn to the levels of these two inputs to assess whether the observed change in composition reflects substitution between capital types or an expansion in overall investment. To this end, we re-estimate equation (2), replacing the dependent variable with the logarithm of the stock of physical capital or the logarithm of the stock of intangible capital. The results are presented in Figure 6.

Before discussing the results, it is important to note that identification in this exercise is weaker than in the analysis of capital composition. Studying input levels, rather than their ratio, requires the standard assumptions of the misallocation literature: comparable production technologies and stable input price levels across firms.¹³ In contrast, our previous analysis of the capital-to-intangible ratio relied only on relative price stability and did not require such assumptions. We therefore interpret the following results as suggestive evidence on the direction of adjustment, rather than as precise estimates of changes in physical or intangible quantities.

Following the introduction of the investment subsidy, treated firms show no significant change in the level of physical capital relative to control firms. By contrast, the stock of intangible capital rises markedly: between 2014 and 2018, treated firms increase their intangible capital by 11 percent—equivalent to roughly 51,000 euros—relative to control firms. This finding confirms that the decline in the capital-to-intangible ratio is driven by an expansion in intangible capital rather than by a contraction in physical capital.

The results in Figure 6 also display mild pre-trends in both types of capital, likely

¹³In particular, we must assume that price levels and output elasticities are similar across treated and control firms, so that changes in the measured capital stock reflect changes in quantities rather than valuation effects.

reflecting common shocks such as asset price inflation, productivity growth, or demand fluctuations. Analyzing the ratio of physical to intangible capital, as in our baseline specification, helps mitigate these confounding factors by differencing out common trends and isolating changes in relative input use. This distinction highlights why the ratio-based analysis provides a cleaner measure of the underlying wedge adjustments: it requires no assumptions about price levels, only about relative prices.

Finally, Online Appendix Figure B.5 explores regional heterogeneity in capital stock responses. We find no change in the stock of physical capital in either high- or low-subsidy regions. In contrast, treated firms in high-subsidy regions exhibit a clear increase in intangible capital relative to control firms, while treated firms in low-subsidy regions show no significant change. This pattern reinforces our earlier conclusion that substantial subsidies are necessary to induce adjustments in capital accumulation—and that these adjustments are concentrated in intangible assets.

3.5 The Extensive Margin of Adjustment

The analysis so far has combined both extensive and intensive margins of adjustment. By using the logarithm of the ratio of physical to intangible capital as the outcome variable, the estimated effects reflect two components: (i) the average change in capital mix among firms that adjust, and (ii) the share of firms that adjust. A large literature on capital adjustment frictions (Caballero et al., 1995; Khan and Thomas, 2008; Eberly et al., 2012; Ottonello and Winberry, 2023) emphasizes the importance of the extensive margin. We now isolate that margin in our setting.

Before proceeding, it is important to note that identification in this exercise is weaker than in our baseline specification. Studying the likelihood of adjustment requires stronger assumptions about price levels and production technologies across firms, similar to those in the misallocation literature. The results should therefore be interpreted as descriptive evidence on adjustment behavior rather than as precise measures of underlying wedges.

We estimate the following event–study specification:

$$\begin{aligned} \text{Adjusted}_{i,t}^h = & \alpha_i + \lambda_t + \beta X_{i,t} \\ & + \sum_{m=-5, m \neq -1}^4 \delta_m \mathbf{1}\{i \in \text{Treated}\} \mathbf{1}\{t = 2015 + m\} + \varepsilon_{i,t}, \end{aligned} \quad (3)$$

where the dependent variable $\text{Adjusted}_{i,t}^h$ equals one if firm i adjusts its stock of capital type $h \in \{\text{Physical}, \text{Intangible}\}$ in year t , and zero otherwise. We define adjustment as an

absolute annual change in the capital stock exceeding one percent:

$$\text{Adjusted}_{i,t}^h = \mathbf{1} \left\{ \left| \frac{K_{i,t+1}^h - K_{i,t}^h}{K_{i,t}^h} \right| > 0.01 \right\}.$$

As before, we include firm and year fixed effects and a vector of time-varying controls. Standard errors are clustered at the firm level. The results are shown in Figure 7.

Following the introduction of the investment subsidy, treated firms are no more likely than control firms to adjust their stock of physical capital. In contrast, they are significantly more likely to adjust their stock of intangible capital. Between 2014 and 2019, the probability of adjusting intangible capital increases by 1.9 percentage points for treated firms relative to control firms—an increase of 2.6 percent relative to the baseline adjustment rate. This pattern is consistent with models of lumpy investment with fixed adjustment costs (Caballero et al., 1995; Khan and Thomas, 2008), in which only sufficiently large shocks trigger reoptimization.

The response along the extensive margin is heterogeneous across firms. As shown in Figure B.6, smaller firms are significantly more likely than larger firms to adjust their stock of intangible capital following the subsidy—consistent with the view that financially constrained firms face higher inaction thresholds and respond more strongly when those constraints are relaxed.

In sum, the subsidy increases the incidence of adjustments in intangible capital—especially among smaller, more constrained firms—while leaving physical capital adjustment rates unchanged. Together with our levels evidence (no change in physical capital, clear growth in intangibles), this indicates that the observed decline in the capital-to-intangible ratio reflects an expansion in intangible investment rather than a contraction of physical capital, without apportioning the total effect across margins.

3.6 Firm Performance

We have shown that the investment tax subsidy led firms to increase investment in intangible assets and shift their capital mix away from physical capital. Our working hypothesis is that, prior to the subsidy, a wedge between the marginal rate of technical substitution and the price ratio distorted capital allocation—specifically, that the marginal return to intangible capital was inefficiently high. If this wedge is reduced by the policy, and firms increase investment in high-return intangible assets, firm performance should improve.

To test this implication, we estimate the following difference-in-differences specifica-

tion:

$$\log Y_{i,t} = \lambda_t + \alpha_i + \beta X_{i,t} + \delta \cdot \mathbf{1}\{i \in \text{Treated}\} \cdot \mathbf{1}\{t \geq 2015\} + \varepsilon_{i,t}, \quad (4)$$

where the outcome variable $Y_{i,t}$ is one of five firm-level performance indicators: (1) cash holdings, (2) short-term debt, (3) long-term debt, (4) operating profits (defined as sales minus operating costs), and (5) an indicator for making strictly positive operating profits. The regression includes firm fixed effects, year fixed effects, and a vector of time-varying firm controls. Standard errors are clustered at the firm level. The coefficient of interest, δ , captures the average treatment effect of the investment tax subsidy. The results are reported in Table IV.

Treated firms increase their cash holdings by 2 percent relative to control firms, corresponding to an average increase of nearly 6,000 euros. This suggests that firms do not immediately invest the full amount of the subsidy; instead, they partially retain it. This behavior is consistent with models of investment subject to fixed adjustment costs, where firms stockpile liquidity in anticipation of future investment episodes. It also aligns with the pecking order theory of Myers and Majluf (1984), in which firms prioritize internal financing.

Firms also adjust their debt structure. Treated firms reduce their short-term debt by 7 percent (approximately 45,000 euros), while increasing long-term debt by 6 percent (around 83,500 euros) relative to control firms. Although total debt increases, its maturity shifts toward the long term. This is consistent with firms becoming more forward-looking and facing looser financial constraints after the policy.

Finally, firm performance improves. Operating profits increase by 1.4 percent relative to control firms—equivalent to a gain of 22,475 euros compared to the unconditional mean. The likelihood of generating strictly positive operating profits also rises: treated firms are 0.2 percentage points more likely to be profitable, a 0.2 percent increase relative to control firms.

Taken together, these results support the interpretation that the investment tax subsidy reduced capital misallocation and improved firm outcomes, both through investment and balance sheet channels.

4 Effects on the Distribution of the Capital-to-Intangible Ratio

The event-study analysis in Section 3 focused on average treatment effects, capturing how the typical firm adjusted its capital mix in response to the investment subsidy. While informative, those results mask the underlying heterogeneity in firms' responses — heterogeneity that is central to understanding how financial frictions shape input choices. This section moves beyond averages to study how the entire distribution of the capital-to-intangible ratio evolves after the policy. By tracing the full distribution rather than its mean, we can assess whether the subsidy primarily affected constrained firms, whether distortions compressed across the economy, and ultimately, how much misallocation the policy alleviated.

4.1 Empirical Approach: The Change-in-Changes Design

So far, we have shown that, on average, firms exposed to the investment subsidy shift their input mix toward intangible assets. We interpret this reallocation as the result of financial frictions being relaxed by the liquidity shock created by the subsidy. Yet average effects can conceal substantial heterogeneity. For instance, if the decline in the mean capital-to-intangible ratio were driven entirely by firms already at the lower end of the distribution, this would be inconsistent with our interpretation — since such firms are less likely to be financially constrained. Understanding how the entire distribution shifts is therefore essential.

To study distributional effects, we use the *Change-in-Changes* (CiC) framework of [Athey and Imbens \(2006\)](#) and [Chernozhukov et al. \(2013\)](#). The CiC estimator extends the logic of difference-in-differences from means to distributions. In standard difference-in-differences, identification relies on the assumption that, absent treatment, the *average* outcome for treated firms would have evolved in parallel with that of control firms. Under this assumption, we can construct a counterfactual path for the mean and obtain the average treatment effect. CiC makes an analogous assumption for the *entire distribution*: absent treatment, the *rank* of each treated firm within its outcome distribution would evolve through the same monotone transformation observed among control firms. This assumption — known as *monotone rank preservation* — allows us to recover counterfactual distributions and estimate *quantile treatment effects* (QTEs) conditional on a firm's initial rank in the outcome distribution.

Formally, for each percentile τ in the initial distribution of outcomes (the distribution

of the logarithm of the capital-to-intangible ratio for treated firms in 2014), we compute the QTE as

$$\text{QTE}_t(\tau) = Q_t^T(\tau) - \underbrace{h_t(Q_{2014}^T(\tau))}_{\text{counterfactual}}, \quad (5)$$

where $h_t(\cdot)$ is a monotone trend mapping defined implicitly by

$$F_t^C(y) = F_{2014}^C(h_t^{-1}(y)),$$

and Q denotes the inverse quantile function. Intuitively, $h_t(\cdot)$ describes how each point in the control group's outcome distribution moves over time. The first step is to estimate this monotone transformation from the control group. The second step assumes that, in the absence of treatment, treated firms would have followed the same transformation. The third step compares the observed outcome of treated firms at each quantile to this counterfactual to obtain $\text{QTE}_t(\tau)$.

The identifying assumptions behind CiC are stronger than those required for mean difference-in-differences. In addition to monotone rank preservation, CiC requires a stronger version of the common-trend assumption — namely, that the untreated outcome distributions for treated and control firms evolve through the same rank-preserving transformation. In our context, this means that, absent the subsidy, any evolution in the capital-to-intangible ratio would have shifted the entire distribution of treated firms in the same way as that of controls.

The intuition is straightforward. In difference-in-differences, we assume that treated and control firms would have experienced the same *shift* in their mean outcomes. In CiC, we instead assume that they would have experienced the same *reshaping* of their entire distributions. This assumption rules out systematic differences in the distribution of idiosyncratic shocks or cyclical exposures that could change firms' relative positions in the outcome distribution. For example, if treated firms drew idiosyncratic shocks from a different distribution, their relative ranks in the capital-to-intangible ratio could change even in the absence of treatment, violating rank preservation. In such cases, mean-independence would still hold and a difference-in-differences design would correctly estimate the average treatment effect, but ranks would not be stable, and CiC would incorrectly attribute these distributional shifts to the policy. However, in most models, relative prices are identical across firms — or at least drawn from the same distribution — making the CiC assumptions plausible in our setting.

For this reason, the quantile treatment effects we estimate should not be interpreted as

structural parameters. However, the overall *shape* of the QTE distribution remains highly informative. If the policy primarily relaxes financial constraints, we should observe the largest effects among firms initially at the top of the capital-to-intangible distribution — those with the highest distortions and strongest reallocation potential. The precise magnitudes should be interpreted with caution, but the qualitative pattern of heterogeneity provides an important test of the underlying mechanism.

4.2 Distributional Effects: Change-in-Changes Results

To examine how the distribution of the logarithm of the capital-to-intangible ratio evolves after the policy, we apply the CiC estimator for all years $t \geq 2015$, using 2014 as the reference year. Inference is obtained through a nonparametric block bootstrap that resamples firms—rather than individual observations—with replacement.¹⁴ We focus on 2017 as a representative post-treatment year, by which time the effects of the policy are fully reflected in firms’ balance sheets but the program remains in force. The results of this estimation are shown in Figure 8.

The quantile treatment effects display a clear and economically meaningful pattern consistent with the presence of financial frictions. Firms in the lower percentiles of the log capital-to-intangible ratio distribution—those already close to the frictionless input mix—show no statistically significant change following the introduction of the subsidy. In contrast, firms in the upper part of the distribution, which are further from the unconstrained input mix and thus more likely to be financially constrained, exhibit a substantial decline in the ratio. For instance, a firm initially at the 75th percentile experiences a roughly 12 percent reduction in its capital-to-intangible ratio relative to a comparable control firm.¹⁵ This heterogeneous response across the distribution reinforces our earlier conclusion: the policy primarily relaxed financing constraints rather than altering relative input prices.

The downward-sloping pattern of the quantile treatment effects implies a compression in the distribution of wedges. Under our identifying assumptions, changes in the capital-to-intangible ratio can be interpreted as changes in the wedge between a firm’s observed

¹⁴Because treatment is assigned at the firm level and outcomes are observed over time, resampling entire firm histories preserves the within-firm serial dependence structure of the data. This procedure is conceptually analogous to clustering standard errors at the firm level: by treating each firm as the resampling unit, we allow for arbitrary autocorrelation and heteroskedasticity within firms while maintaining independence across firms. This approach ensures inference that is robust to serial dependence, which is essential in event-study and CiC settings.

¹⁵Results for all post-treatment years are reported in Online Appendix Figure B.7. The same qualitative pattern—null effects for low percentiles and negative effects for high percentiles—persists throughout the sample and becomes more pronounced over time.

and frictionless input mix. The investment subsidy thus reduced overall distortions by narrowing the spread of wedges across firms. Most of this compression arises from a reduction in wedges among firms in the upper tail of the distribution—those that were initially the most constrained. While these effects should be interpreted cautiously, they suggest that the policy improved allocative efficiency by bringing the most distorted firms closer to their unconstrained equilibrium.

To visualize this compression directly, we compute the gap between the post-treatment and counterfactual distributions implied by the CiC estimator. For each percentile τ , the CDF gap is defined as

$$G_t(y) = \mathbb{P}(Y_t^T \leq y) - \mathbb{P}(Y_t^{\text{CF}} \leq y),$$

which measures the difference between the empirical distribution of treated firms in year $t \geq 2015$ and the counterfactual distribution predicted in the absence of treatment. Figure 9 plots this gap for the post-treatment period.

The CDF gap provides a complementary visualization to the QTE plot. While the QTEs capture how treatment effects vary across percentiles, the CDF gap shows how probability mass shifts across the entire distribution. The gap is negative at low values and positive at high values, with a single crossing point—indicating that mass shifted from the upper tail toward the center. In economic terms, there are fewer firms with large wedges and more firms with moderate ones. This pattern confirms that the subsidy compressed the wedge distribution, primarily benefiting the most constrained firms. The single crossing further suggests a first-order stochastic dominance improvement, consistent with a more efficient allocation of inputs across firms.¹⁶

Having documented these distributional shifts, we next translate the estimated quantile-level effects into a distribution of wedges that rationalizes the data under our model.

4.3 Recovering the Distribution of Wedges

In Section 3, we used the average treatment effect to compute a lower bound on the average wedge in the economy prior to the introduction of the investment subsidy. We now go further and recover the entire *distribution* of wedges for the least distorted economy that is consistent with our empirical results.¹⁷

¹⁶This distributional compression mirrors the aggregate decline in the average wedge documented earlier, reinforcing the interpretation that the policy alleviated financing distortions rather than altering relative prices.

¹⁷We define the *least distorted economy* as the economy in which, for each firm, the wedge equals the lower bound implied by the quantile treatment effects reported in Figure 8.

Intuitively, our goal is to characterize the minimum degree of misallocation compatible with the data. Because the wedge ω is bounded below by one, we can construct for each percentile τ a lower bound $\tilde{\omega}(\tau)$ on the wedge as

$$\tilde{\omega}(\tau) = \max \left\{ 1, \exp \left\{ - \min_{t \geq 2015} \text{QTE}_t(\tau) \right\} \right\}. \quad (6)$$

For any percentile τ in which the quantile treatment effect is negative for all post-treatment years, the minimum of $\text{QTE}_t(\tau)$ corresponds to the largest decline in the wedge experienced by firms initially in that percentile. Because $\omega \geq 1$, this minimum identifies a sharper lower bound than the trivial value of one.¹⁸ Conversely, for percentiles where the quantile treatment effect is non-negative, the lower bound remains one, as the data do not provide additional information about reductions in distortions.

Equation (6) provides a lower bound for each percentile τ . To transform this mapping into a full distribution of wedges, we exploit the fact that the percentile rank τ of the underlying variable (the logarithm of the capital-to-intangible ratio) is uniformly distributed on the unit interval.¹⁹ We observe $\tilde{\omega}(\tau)$ on a fine grid of percentiles and linearly interpolate between grid points to approximate the continuous function. Recovering the distribution $F^*(\tilde{\omega})$ of wedges from the estimated quantile curve is a standard change-of-variables problem—closely related to constructing distributions from quantile functions (Serfling, 2009). Full details of the algorithm are provided in Appendix D.

We use this algorithm to compute the distribution $F^*(\tilde{\omega})$ of the minimum wedges that rationalize the data under our theoretical structure. By construction, this distribution represents the least distorted economy pointwise—firm by firm—because it takes the minimum wedge consistent with the evidence. The resulting distribution is shown in Figure 10.

The recovered distribution features a discrete mass at $\omega = 1$, the lower bound of the support. Approximately five percent of firms are located at this point, implying that they are effectively unconstrained. Hence, at most five percent of firms in our sample operate without financial frictions, suggesting that the Portuguese corporate sector is substantially distorted in terms of the marginal rate of technical substitution. The median wedge is 1.13, almost identical to the lower bound for the average wedge reported in Section 3. Moreover, distortions are quantitatively significant: twenty-five percent of firms exhibit wedges of at least 1.2, meaning their capital-to-intangible ratios are at least twenty percent

¹⁸Since the outcome variable is expressed in logs, the transformation from log changes to level changes is exponential.

¹⁹This follows directly from the probability integral transform (Feller, 1991).

higher than the frictionless benchmark.

The distribution $F^*(\tilde{\omega})$ therefore provides the tightest possible lower bound on misallocation consistent with the data. Any other distribution supported on the same domain that first-order stochastically dominates $F^*(\tilde{\omega})$ is also consistent with the data and the model.²⁰ In this sense, the economy characterized by $F^*(\tilde{\omega})$ is the *least distorted* allocation compatible with our results. Any empirically or theoretically plausible alternative economy must be weakly more distorted—that is, must first-order stochastically dominate $F^*(\tilde{\omega})$. This construction transforms the local evidence from the investment subsidy into a global statement about the minimum degree of misallocation in the economy.

This exercise also connects our analysis to two broader literatures. First, it provides a micro–empirical bridge to the misallocation literature (Hsieh and Klenow, 2009), which typically infers wedges indirectly from output and input shares under strong assumptions about technology and prices. In contrast, our approach recovers the entire distribution of wedges directly from observable firm behavior, requiring minimal structural assumptions. Second, it relates naturally to the wedge–accounting tradition in macroeconomics (Chari, Kehoe and McGrattan, 2007), which interprets aggregate inefficiencies as arising from implicit distortions in firms’ first-order conditions. Our results can thus be viewed as a micro–level analogue of this framework: we recover the empirical distribution of wedges that rationalizes firm–level choices, quantify its dispersion, and identify the least distorted allocation consistent with the data. By doing so, we provide a direct measure of the magnitude and distributional shape of financial distortions in the economy.

Finally, these findings complete the narrative begun in the event–study analysis. The average decline in the capital-to-intangible ratio reflected an economy-wide easing of financial constraints; the distributional evidence now reveals that this adjustment was concentrated among the most distorted firms. By recovering the least distorted distribution of wedges, we quantify not only how much the policy improved efficiency on average, but also how it reshaped the dispersion of distortions across firms.

²⁰To see this, suppose that wedges are instead given by $\omega' = \omega + \eta$, where $\eta \geq 0$ a.s.. The new set of wedges is weakly larger almost surely, which represents a more constrained economy. Moreover, it follows that

$$F_{\omega'}(x) = \mathbb{P}(\omega + \eta \leq x) = \mathbb{P}(\omega \leq x - \eta) \leq \mathbb{P}(\omega \leq x) = F^*(x),$$

and so $F_{\omega'}$ first order stochastically dominates F^* .

5 Conclusion

This paper shows that financial frictions distort not only the allocation of resources across firms but also the composition of investment within firms. Exploiting a quasi-experimental policy that increased firms' liquidity without changing relative prices, we find that treated firms rebalanced their input mix toward intangible capital. This response is inconsistent with a frictionless model but aligns with a framework in which financing constraints raise the effective cost of intangible investment. By linking a well-identified empirical setting to a simple theoretical structure, we provide direct evidence that financial frictions shape firms' capital composition.

Our results indicate that the investment subsidy reduced the wedge between the marginal rate of technical substitution and the relative price of inputs by about eleven percent, relaxing financial constraints and improving the efficiency of capital allocation. Heterogeneity across regions and firms supports this interpretation: effects are strongest in areas with more generous subsidies and among smaller firms, which are more likely to be constrained. The policy also improved firms' balance sheets and performance, leading to higher liquidity, longer debt maturity, and higher operating profits.

Beyond the average effects, we show that the policy compressed the distribution of wedges, reducing distortions primarily among firms that were initially the most constrained. Using a Change-in-Changes framework, we recover quantile treatment effects along the entire distribution of the capital-to-intangible ratio, revealing that adjustment is concentrated in the upper tail. Building on these results, we propose a new method to recover the full distribution of wedges for the least distorted economy consistent with the data. The recovered distribution provides a lower bound on misallocation: at most five percent of firms appear unconstrained, and one-quarter exhibit wedges of at least twenty percent.

Together, these findings contribute to the literatures on financial frictions, intangible investment, and misallocation. Empirically, we offer causal evidence that financial constraints distort firms' capital mix and that liquidity shocks can alleviate these distortions. Methodologically, we provide a new way to infer the cross-sectional distribution of wedges directly from firm-level responses, bridging the gap between reduced-form evidence and the structural approaches of the misallocation and wedge-accounting literatures. Our framework thus offers a tractable empirical link between firm-level distortions and aggregate productivity losses, providing a foundation for future work quantifying the macroeconomic implications of financial frictions.

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Figures and Tables

FIGURE 1: Distribution of Capital-to-Intangible Ratio

This figure plots the kernel density of the logarithm of the capital-to-intangible ratio for all firms in our sample in 2014. We present the density for control firms and treated firms.

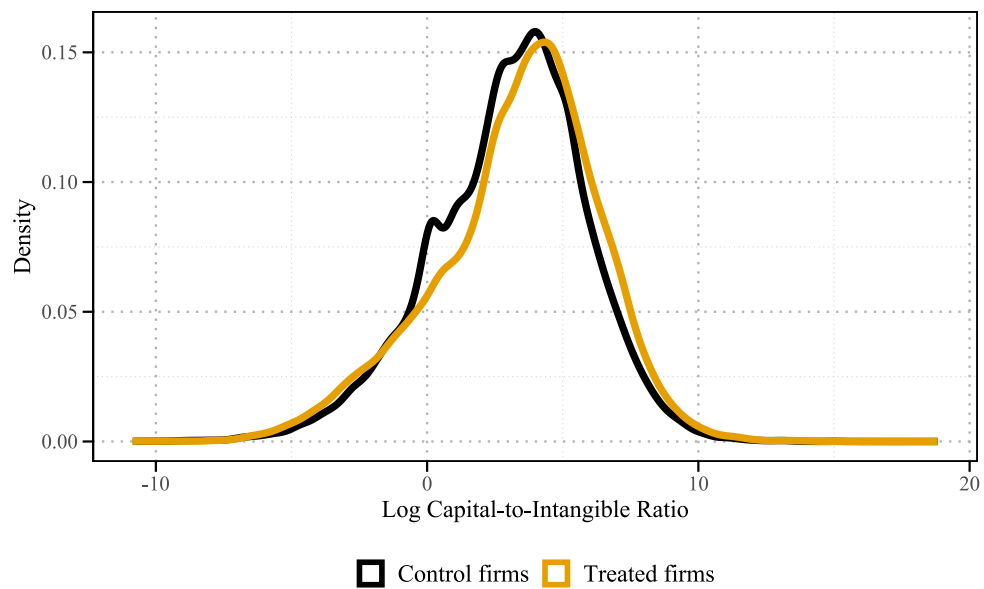


FIGURE 2: Effect on Aggregate Capital Stocks

This figure compares the evolution of the sum of the stock of physical capital and the sum of the stock of intangible capital for treated and control firms. We first aggregate these variables and then take the logarithms and scale by their values in 2014.

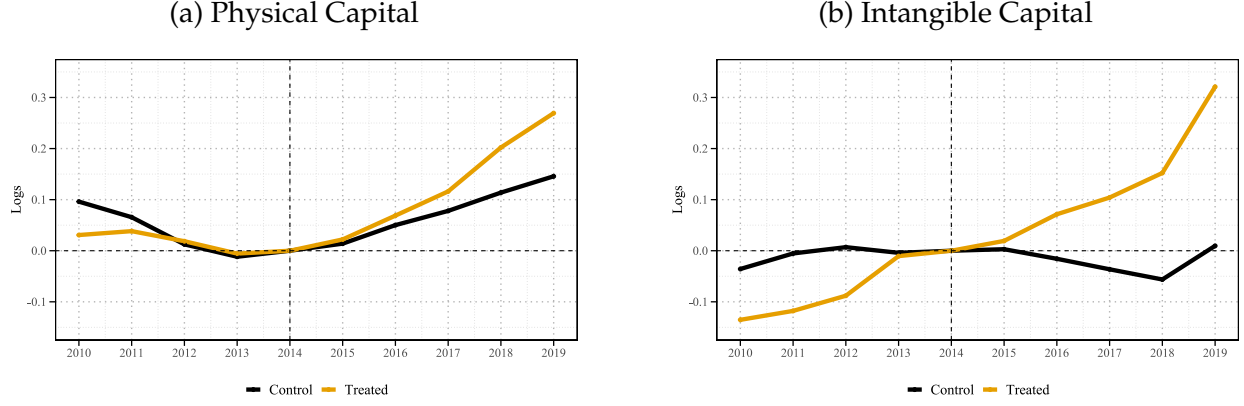


FIGURE 3: Effect on Firm-Level Capital-to-Intangible Ratio

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

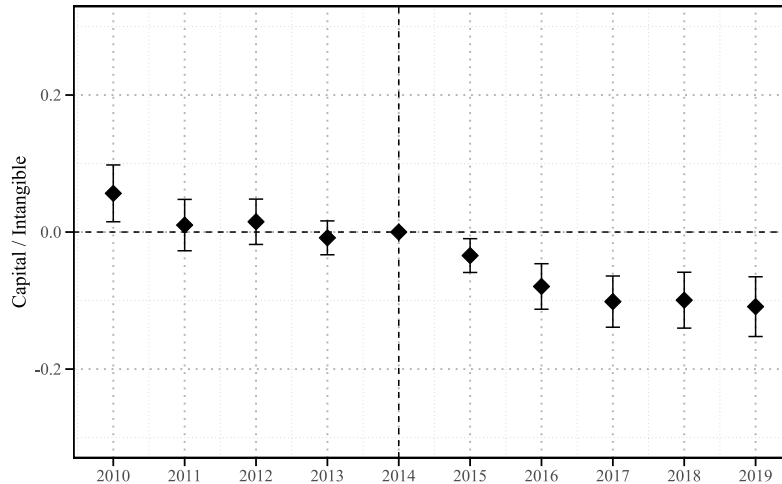


FIGURE 4: Effect on Firm-Level Capital-to-Intangible Ratio - Decomposition by Region

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We present results for three samples: (1) including all firms, (2) including only firms in regions where the investment subsidy was 10 percent, and (3) including only firms in regions where the investment subsidy was 25 percent. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

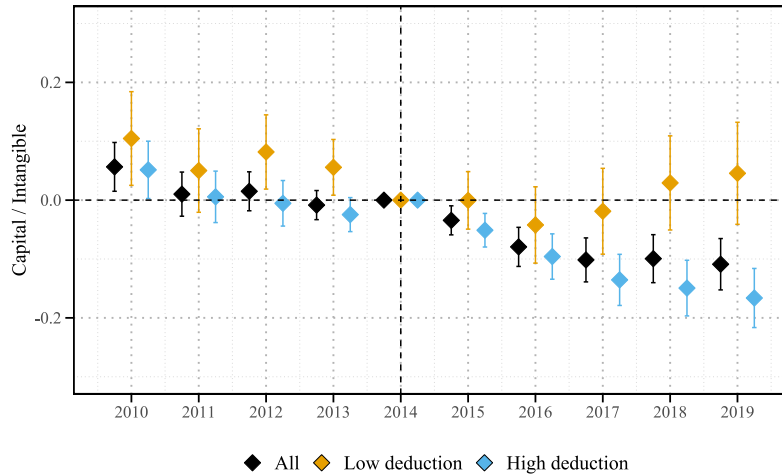


FIGURE 5: Effect on Firm-Level Capital-to-Intangible Ratio - Decomposition by Firm Size

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We split firms according to their level of sales in 2014. Large firms are defined as firms above the cross-sectional median and small firms are defined as firms below the cross-sectional median. We present results for three samples: (1) including all firms, (2) including only small firms, and (3) including only large firms. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

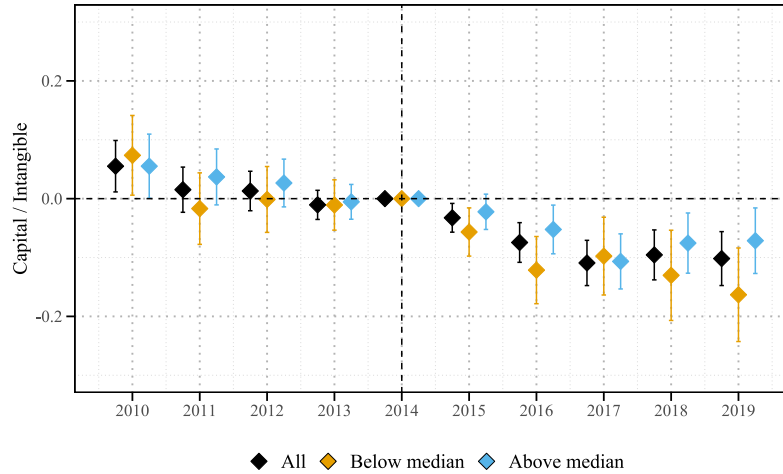


FIGURE 6: Effect on Firm-Level Capital Stocks

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is either the logarithm of the stock of physical capital at the end of the period or the logarithm of the stock of intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

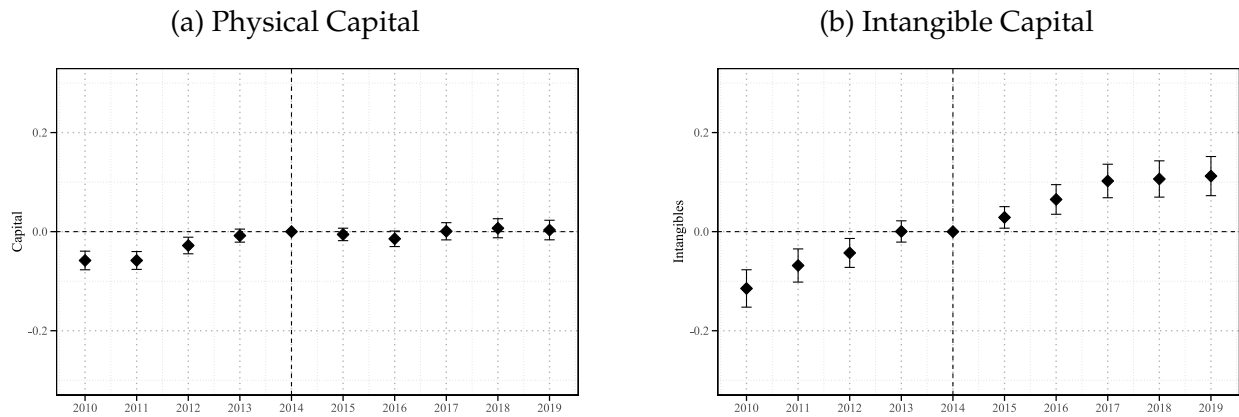


FIGURE 7: Effect on Firm-Level Probability of Adjustment

This Figure presents the results of estimating equation (3) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is either the an indicator that takes the value of 1 if the firm has adjusted its stock of physical capital and zero otherwise or an outcome variable that takes the value of 1 if the firm has adjusted its stock of intangible capital and zero otherwise. We define adjustment as having a change in the stock which is larger in absolute value than 1 percent of the lagged stock. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

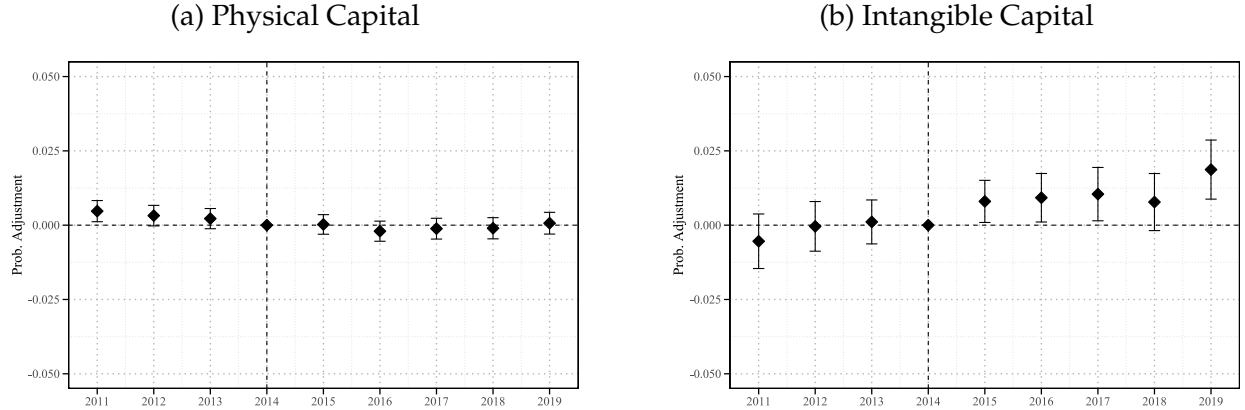


FIGURE 8: Effect on Distribution of Capital-to-Intangible Ratio

This figure presents the results of a Change-in-Changes estimation using a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. For each percentile τ of the distribution of the outcome variable for treated firms in 2014 (the base year), we estimate the quantile treatment effect (QTE) between 2017 and the base year of 2014. We compute standard errors via a nonparametric block bootstrap that resamples firms (rather than individual observations) and present 95 percent confidence intervals.

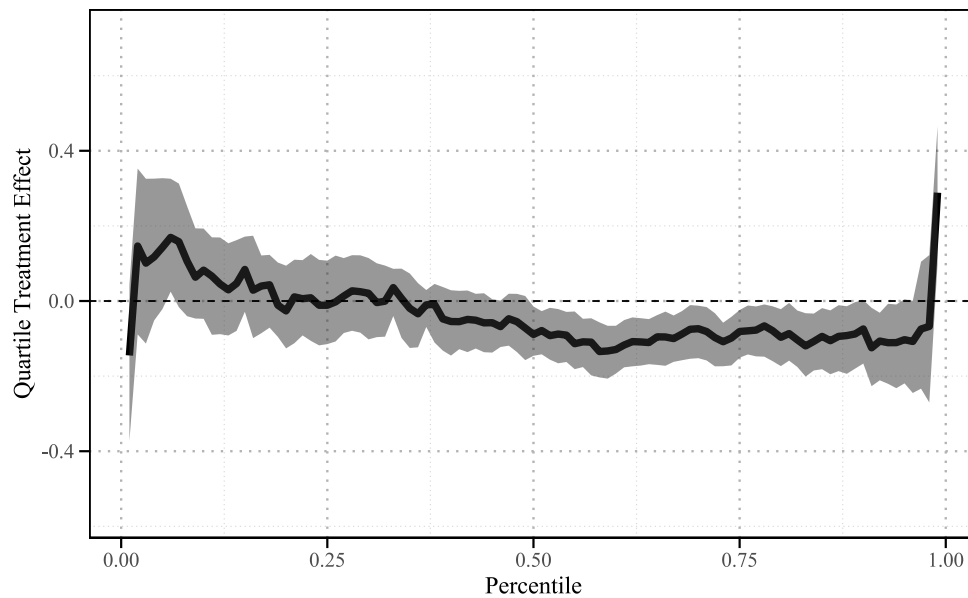


FIGURE 9: Distribution Gap

This figure presents the results of a Change-in-Changes estimation using a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We present the difference between the distribution of outcomes for treated outcomes in 2017 and the counterfactual distribution for the same year computed using the CiC estimator.

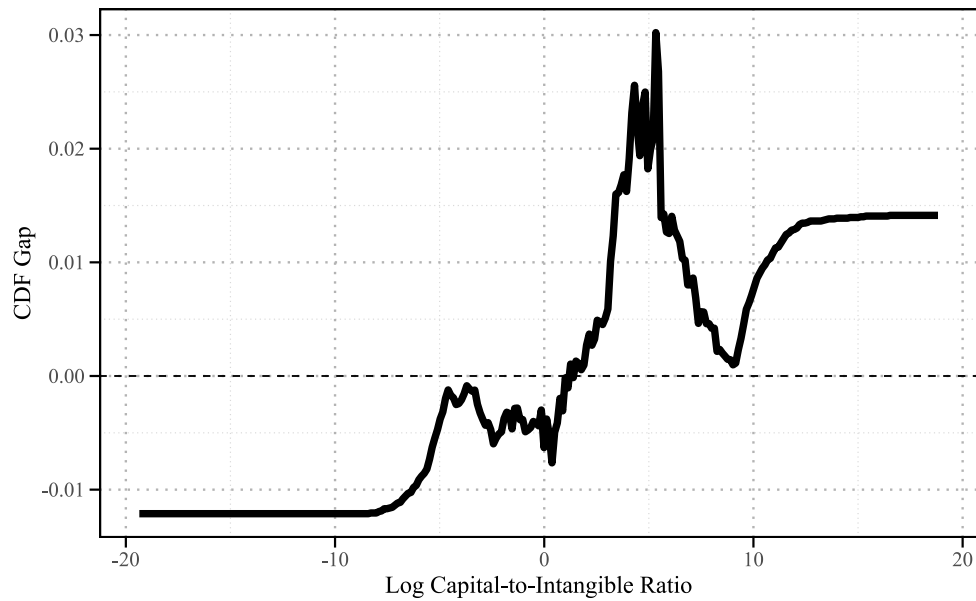


FIGURE 10: Distribution of Wedges in Least Distorted Economy

This figure presents the estimated distribution of wedges in the least distorted economy which is consistent with the empirical evidence.

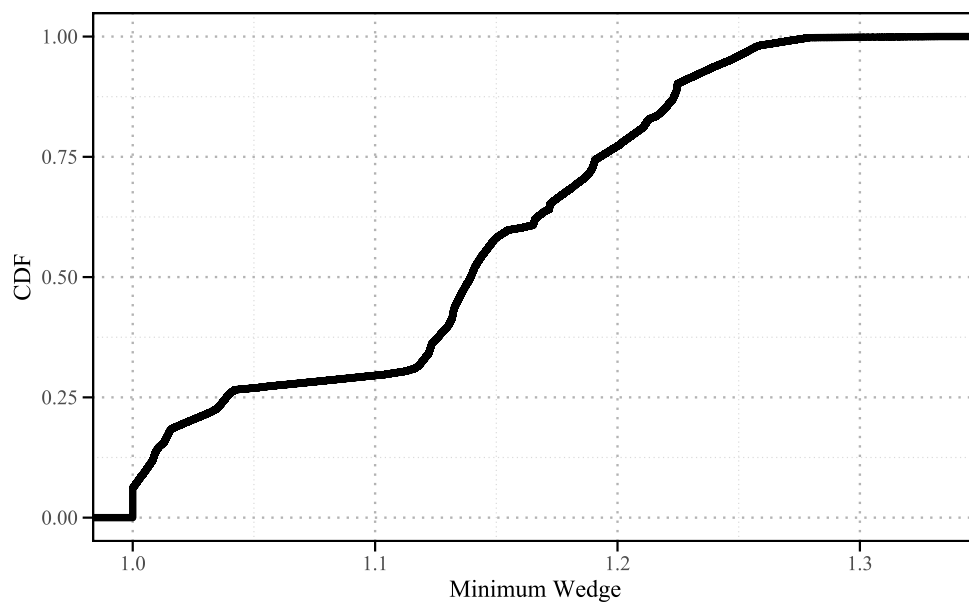


TABLE I: Cash Generated by Operations

This table presents the impact of the investment subsidy on a firm. The firm carries out an investment of 20 in either physical or intangible capital in year 0. The asset fully depreciates in five years, and so the added depreciation is 4. The tax rate is 30 percent and the investment subsidy is 25 percent.

Year:	0	1	2	3	4	5
<i>Income statement</i>						
Sales	60	60	60	60	60	60
Operating costs	30	30	30	30	30	30
Investment costs	20	0	0	0	0	0
Depreciation	5	9	9	9	9	9
<i>Cash flow</i>						
Investment subsidy	5	0	0	0	0	0
Tax benefit from depreciation	1.5	2.7	2.7	2.7	2.7	2.7
Free cash flow from operations	7.5	23.7	23.7	23.7	23.7	23.7

TABLE II: Summary Statistics

This table presents summary statistics for all firms in our sample in 2014. We consider two groups: all firms, firms in the treated group, and firms in the control group. For each group we present the cross-sectional average of: total assets, total sales, the number of workers, the wage bill, the leverage ratio (measured as the ratio of total liabilities to total assets), the ratio of cash to total assets, the ratio of EBITDA to total sales, tangible assets (measured using PP&E), and intangible assets. We also present the difference in means between treated firms and control firms. For each difference, we compute the t-statistic and present the significance level of the difference where ***, **, and * denote significance at the 0.1, 1, or 5 percent level, respectively.

	All firms	Treated	Control	Difference
Assets	1,597,737	1,450,813	1,644,305	-193,392
Sales	987,161	1,248,426	904,350	344,076*
Number of workers	8.13	11.92	6.93	5.00***
Wage bill	139,068	200,572	119,573	80,998***
Leverage ratio	3.30	4.02	3.07	0.94
Cash / Total assets	0.14	0.16	0.13	0.03
EBITDA / Sales	-0.63	-0.68	-0.62	-0.06
Tangible assets	318,396	469,059	270,641	198,414***
Intangible assets	98,442	30,596	119,946	-89,350***
Number of firms	311,731	75,026	236,705	-

TABLE III: Variance Decomposition of Capital-to-Intangible Ratio

This table presents the results of a variance decomposition where the outcome variable is the logarithm of the capital-to-intangible ratio. We use data up to 2014. We present the R^2 of the regression, which represents the share of the variation that is explained by the fixed effects.

	(1)	(2)	(3)	(4)
R^2 (%)	15.86	16.39	92.80	92.80
Sector FE	✓			
Sector-Year FE		✓		
Firm FE			✓	✓
Year FE				✓
Observations	263,018	263,018	263,018	263,018

TABLE IV: Effect on Firm-Level Outcomes

This Table presents the results of estimating equation (4). The outcome variable is one of the following five variables: (1) the logarithm of cash holdings, (2) the logarithm of short-term debt, (3) the logarithm of long-term debt, (4) the logarithm of operating profits, which we define as the difference between sales and operating costs, and (5) an indicator variable that takes the value of one if the firm makes strictly positive operating profits, and zero if otherwise. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effect. We cluster errors at the firm level. ***, **, and * denote significance at the 0.1, 1, or 5 percent level, respectively.

	Cash	ST debt	LT debt	Profits	Positive Profits
Treated $\times$ Post 2014	0.0197* (0.0096)	-0.0671*** (0.0197)	0.0580*** (0.0125)	0.0137*** (0.0028)	0.0019* (0.0008)
Firm FE	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓
Firm controls	✓	✓	✓	✓	✓
Observations	463,486	198,111	300,810	464,838	470,423
R ²	0.89	0.89	0.92	0.99	0.44

Online Appendix

A Summary Statistics

FIGURE A.1: Share of Treated Firms

This Figure presents the the share of treated firms in 2014 among all firms. We consider six quantities: number of workers, total sales, total wage bill, stock of physical capital (measured using PP&E), total assets, and the number of firms.

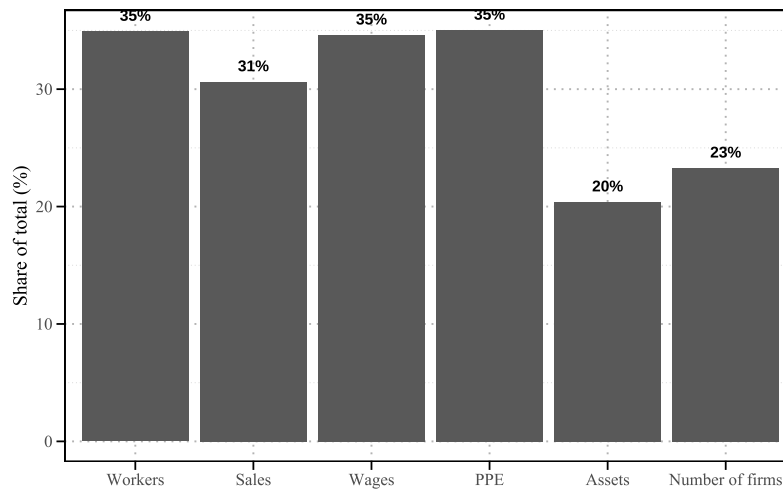


FIGURE A.2: Share of Firms in Regions with a High Deduction

This Figure presents the the share of firms in regions with a high deduction among all treated firms in 2014. We consider six quantities: number of workers, total sales, total wage bill, stock of physical capital (measured using PP&E), total assets, and the number of firms.

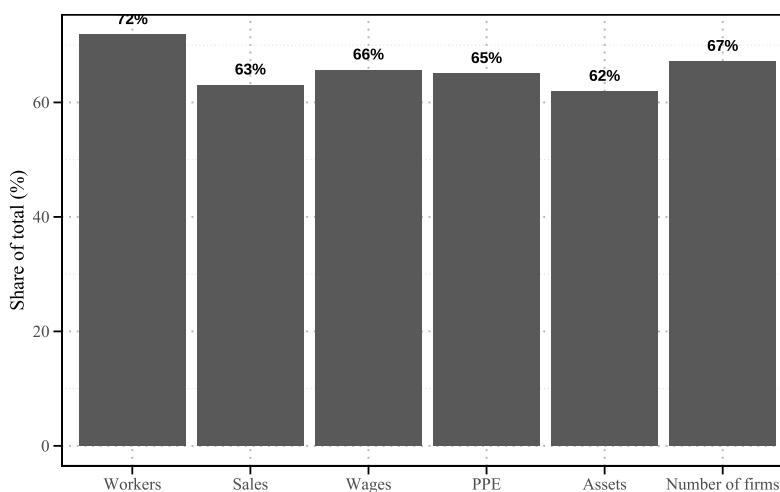


FIGURE A.3: Share of Firms in Regions with a High Deduction

This figure compares the evolution of the aggregate ratio of physical capital to intangible capital for treated and control firms. We first aggregate these variables, compute the ratio, and then take the logarithms and scale by their values in 2014.

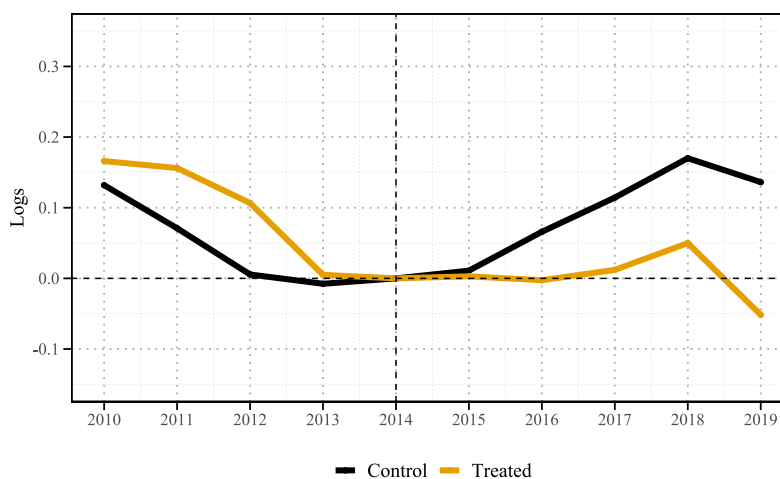


FIGURE A.4: Effect on Aggregate Investment

This figure compares the evolution of the aggregate investment in physical capital and intangible capital for treated and control firms. We first aggregate the stock of both type of capital for both types of firm and then compute the change in logs. We then subtract the level of investment in 2014.

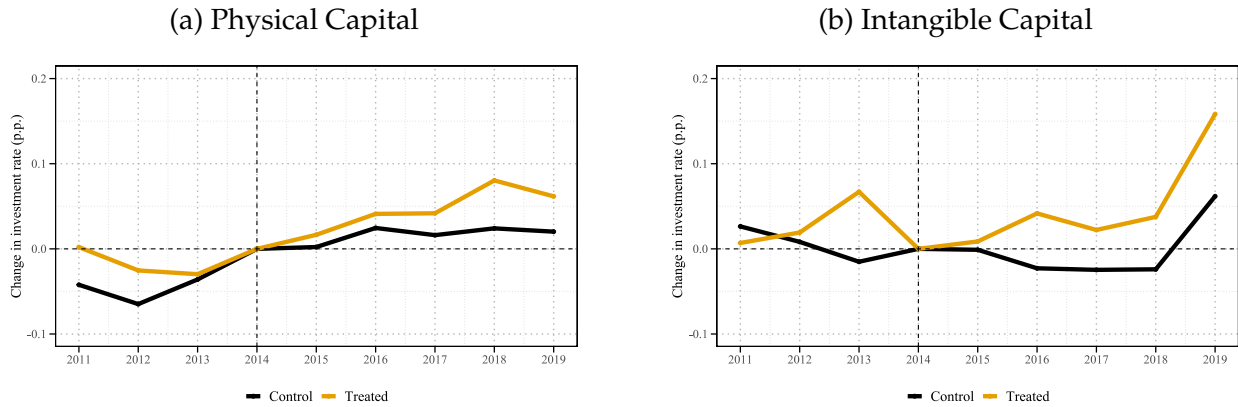


FIGURE A.5: Effect on Aggregate Probability of Adjustment

This figure compares the evolution of the aggregate probability of investing in either physical capital or intangible capital for treated and control firms. For each group of firms, we compute the share of firms that invest more than 1 percent in absolute value, relative to the lagged stock. We then subtract the level of this variable in 2014 to scale the time series.

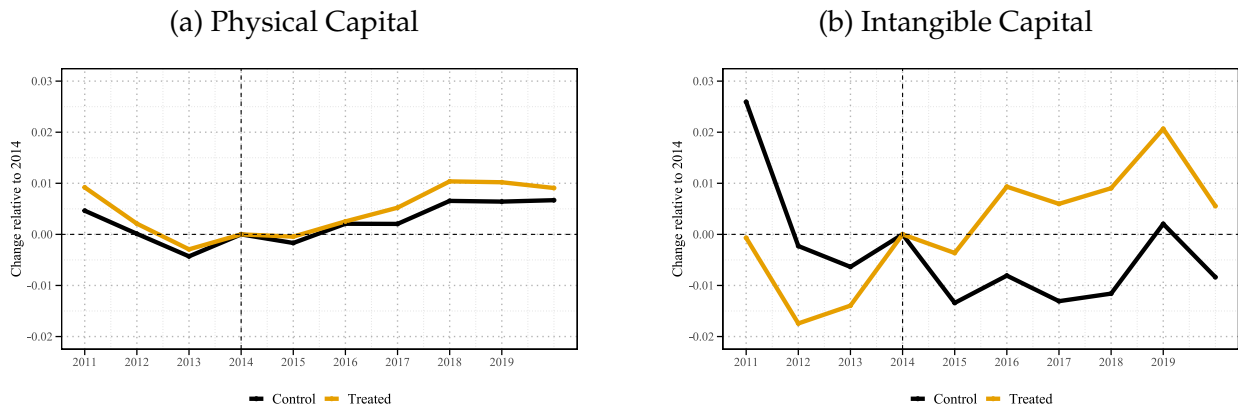


FIGURE A.6: Effect on Different Types of Intangibles

This Figure compares the evolution of different types of intangible assets for treated and control firms. We consider four different types of intangible capital - goodwill, R&D, patents, and other intangibles. For each type of intangible capital, we compute the total stock for treated and control firms. We then take the logarithm and subtract the value on 2014. We present the different between the time series for treated firms and the time series for control firms.

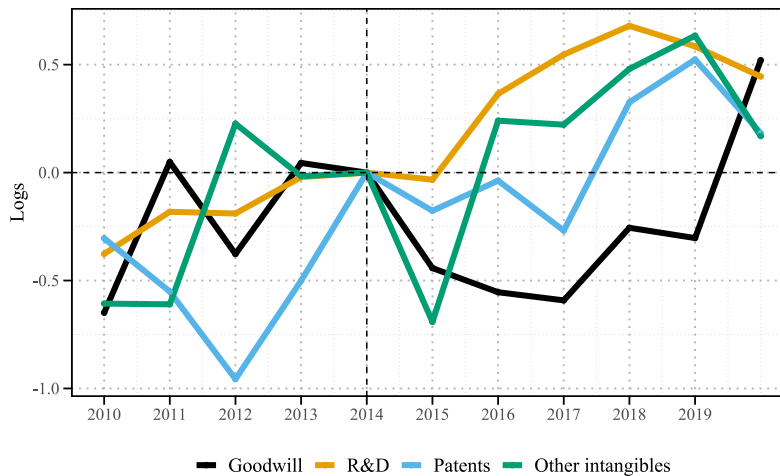


TABLE A.I: Treated Sectors

This table presents the list of treated sectors. For each sector, we also present the share of the number of firms, total assets, sales, or number of workers of that sector relative to all treated firms. All shares are in percentage. The treated sectors are: (1) mining (sector 0500 to 0999), (2) manufacturing (sector 1000 to 3399), (3) hotels & restaurants (sector 5500 to 5699), (4) editing services (sector 5800 to 5899 and sector 5910 to 5919), (5) IT services (sector 6200 to 6299 and sector 6310 to 6319), (6) R&D services (sector 7200 to 7299), and (7) other services (sectors 7721, 9004, 9104, 9311, 9321, 9329, 9329, 9604, 8211, and 8291). All sectors codes use the CAE Rev. 3 classification.

	Number of firms	Assets	Sales	Workers
Mining	0.92	2.11	1.00	0.97
Manufacturing	45.19	71.67	84.37	67.60
Hotels & Restaurants	40.25	16.61	8.27	22.90
Editing & Cinema	3.04	2.41	1.71	1.61
IT services	6.68	3.51	3.71	5.11
R&D	0.58	0.59	0.18	0.46
Other services	3.34	3.11	0.77	1.34

TABLE A.II: Summary Statistics - Treated Firms

This table presents summary statistics for all treated firms in our sample in 2014. We consider two groups: all treated firms, treated firms in regions with a high investment subsidy, and treated firms in regions with a low investment subsidy. For each group we present the cross-sectional average of: total assets, total sales, the number of workers, the wage bill, the leverage ratio (measured as the ratio of total liabilities to total assets), the ratio of cash to total assets, the ratio of EBITDA to total sales, tangible assets (measured using PP&E), and intangible assets. We also present the difference in means between treated firms in regions with a high tax rebate and treated firms in regions with a low tax rebate. For each difference, we compute the t-statistic and present the significance level of the difference where ***, **, and * denote significance at the 0.1, 1, or 5 percent level, respectively.

	All treated firms	High deduction	Low deduction	Difference
Assets	1,450,813	1,351,074	1,659,463	-308,390
Sales	1,248,426	1,168,227	1,416,199	-247,972
Number of workers	11.92	12.68	10.34	2.33***
Wage bill	200,572	194,668	212,921	-18,253
Leverage ratio	4.02	3.06	6.02	-2.97
Cash / Total assets	0.16	0.15	0.19	-0.04***
EBITDA / Sales	-0.68	-0.66	-0.72	0.06
Tangible assets	469,059	451,727	505,319	-53,592
Intangible assets	30,596	20,591	51,527	-30,836*
Number of firms	75,026	24,265	50,761	-

B Additional Firm-Level Results

FIGURE B.1: Effect on Firm-Level Capital-to-Intangible Ratio

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We estimate the regression in three samples: (1) including all firms, (2) including only firms that are not balanced, and (3) including only firms that are balanced. Balanced firms are firms that have observations for all years. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

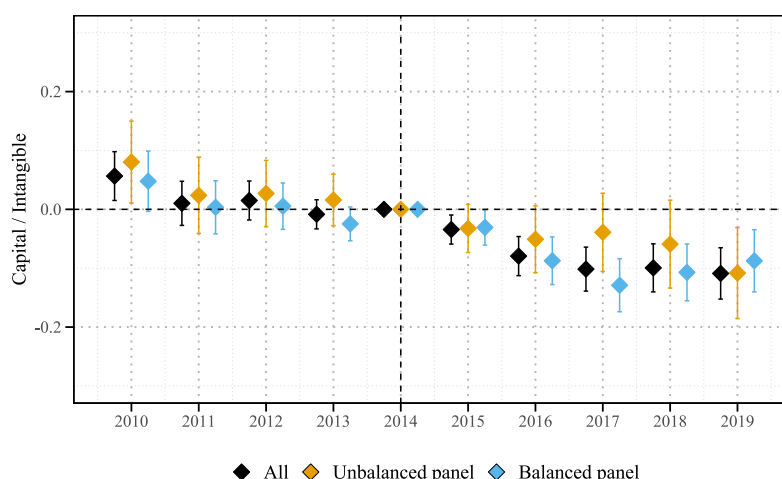


FIGURE B.2: Effect on Firm-Level Capital-to-Intangible Ratio

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We estimate the regression in three samples: (1) including all firms, (2) including only firms that did not have any positive profits before taxes between 2010 and 2014, and (3) including only firms that had at least one year of positive profits before taxes between 2010 and 2014. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

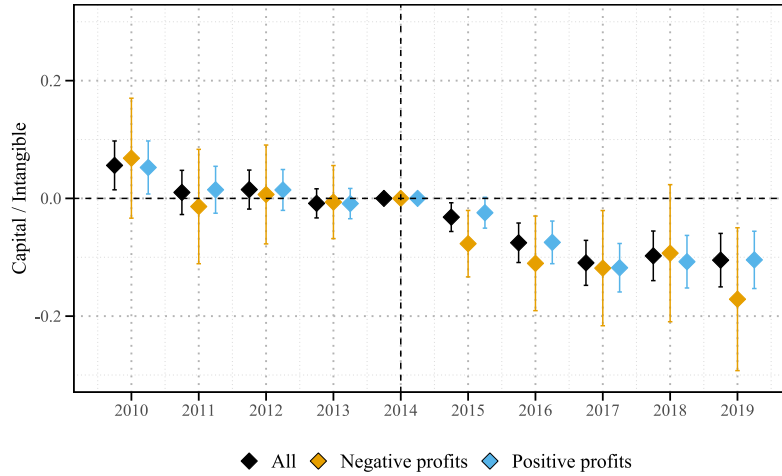


FIGURE B.3: Effect on Firm-Level Capital-to-Intangible Ratio - Decomposition by Firm Size

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We consider two measures of size: total assets and stock of physical capital. We split firms according to the measure of size in 2014. Large firms are defined as firms above the cross-sectional median and small firms are defined as firms below the cross-sectional median. We present results for three samples: (1) including all firms, (2) including only small firms, and (3) including only large firms. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

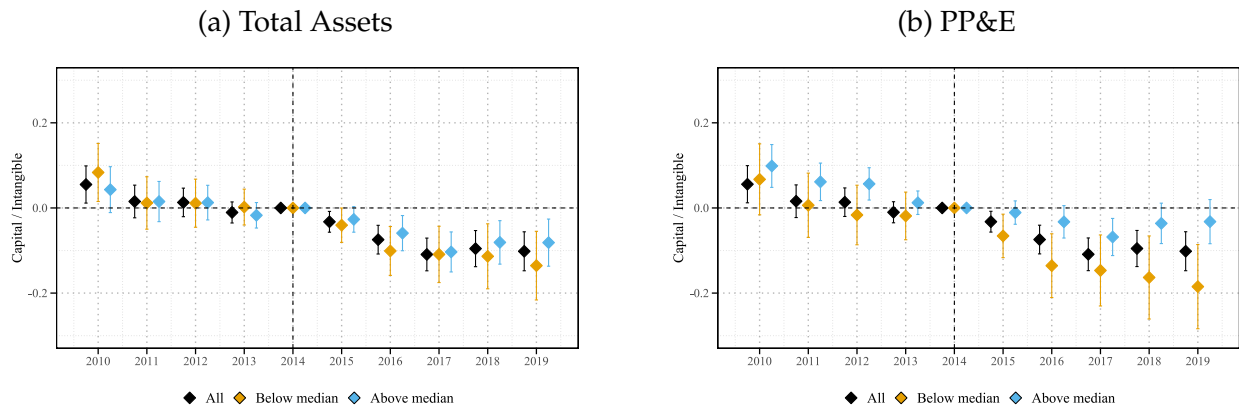


FIGURE B.4: Effect on Distribution of Capital Ratio

This Figure presents the results of estimating equation

$$\mathbf{1}\{p_{i,t+1} = p\} = \lambda_t + \alpha_i + \beta X_{i,t} + \delta^p \cdot \mathbf{1}\{i \in \text{Treated}\} \cdot \mathbf{1}\{t \geq 2015\} + \varepsilon_{i,t},$$

on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is an indicator variable which takes the value of one if the firm's end-of-period capital ratio belongs to a given percentile, and zero if otherwise. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effect where we compare treated vs. control firms. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

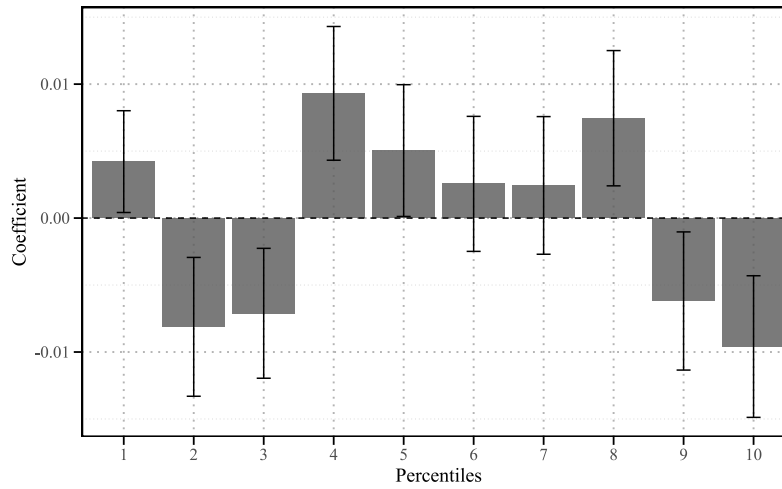


FIGURE B.5: Effect on Firm-Level Capital Stocks - Decomposition by Region

This Figure presents the results of estimating equation (2) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is either the logarithm of the stock of physical capital at the end of the period or the logarithm of the stock of intangible capital at the end of the period. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We present results for three samples: (1) including all firms, (2) including only firms in regions where the investment subsidy was 10 percent, and (3) including only firms in regions where the investment subsidy was 25 percent. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

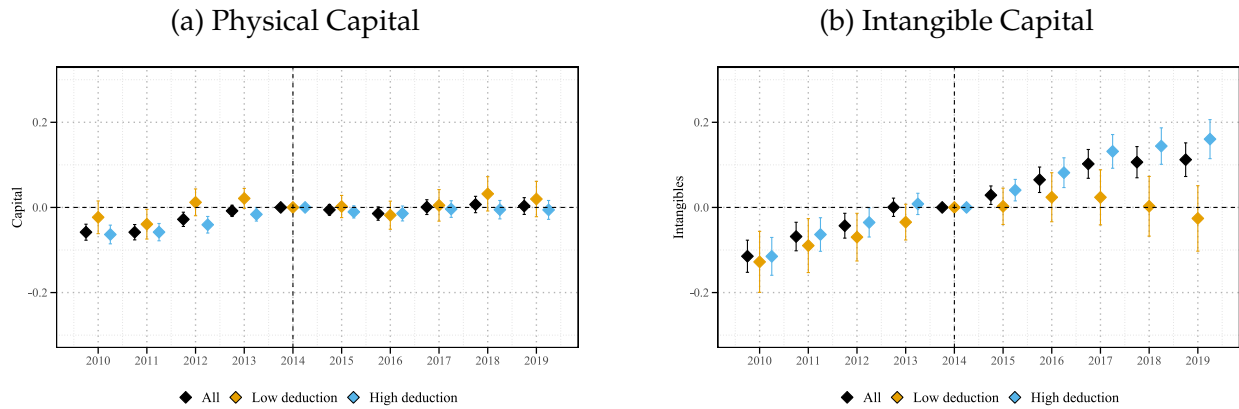


FIGURE B.6: Effect on Firm-Level Probability of Adjustment - Decomposition by Firm Size

This Figure presents the results of estimating equation (3) on a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is either the an indicator that takes the value of 1 if the firm has adjusted its stock of physical capital and zero otherwise or an outcome variable that takes the value of 1 if the firm has adjusted its stock of intangible capital and zero otherwise. We define adjustment as having a change in the stock which is larger in absolute value than 1 percent of the lagged stock. We include firm and year fixed effects, as well as a vector of time-varying firm controls: logarithm of sales, the logarithm of total assets, the logarithm of the number of workers, leverage, the ratio of cash holdings to total assets, the ratio of EBITDA to total sales, the ratio of physical capital to total assets, and the logarithm of total sectoral sales. We present the average treatment effects over time, where we compare treated vs. control firms, using 2014 as the base year. Therefore, the coefficient for the year $2014 + m$ can be interpreted as the average treatment effect on the outcome variable between 2014 and $2014 + m$. We split firms according to their level of sales in 2014. Large firms are defined as firms above the cross-sectional median and small firms are defined as firms below the cross-sectional median. We present results for three samples: (1) including all firms, (2) including only small firms, and (3) including only large firms. We cluster errors at the firm level and present both the coefficients and a 95 percent confidence interval.

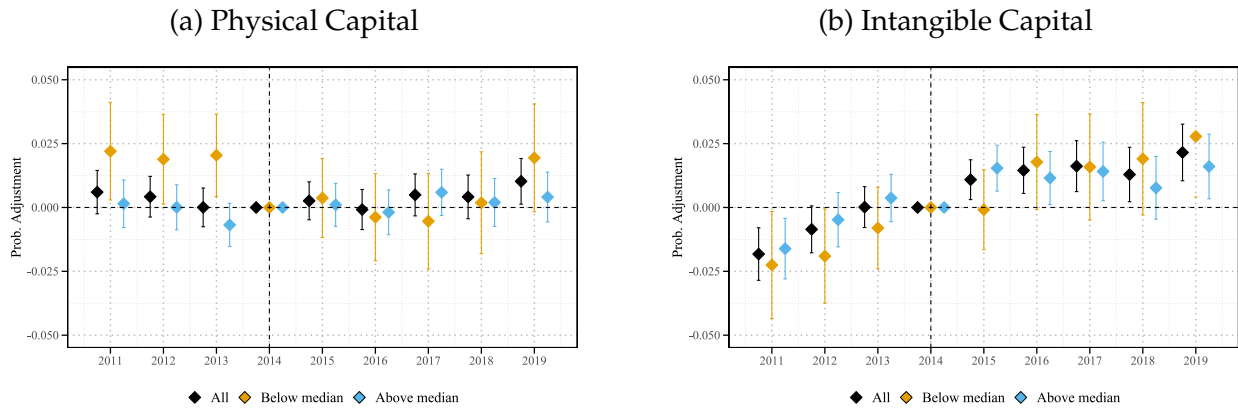
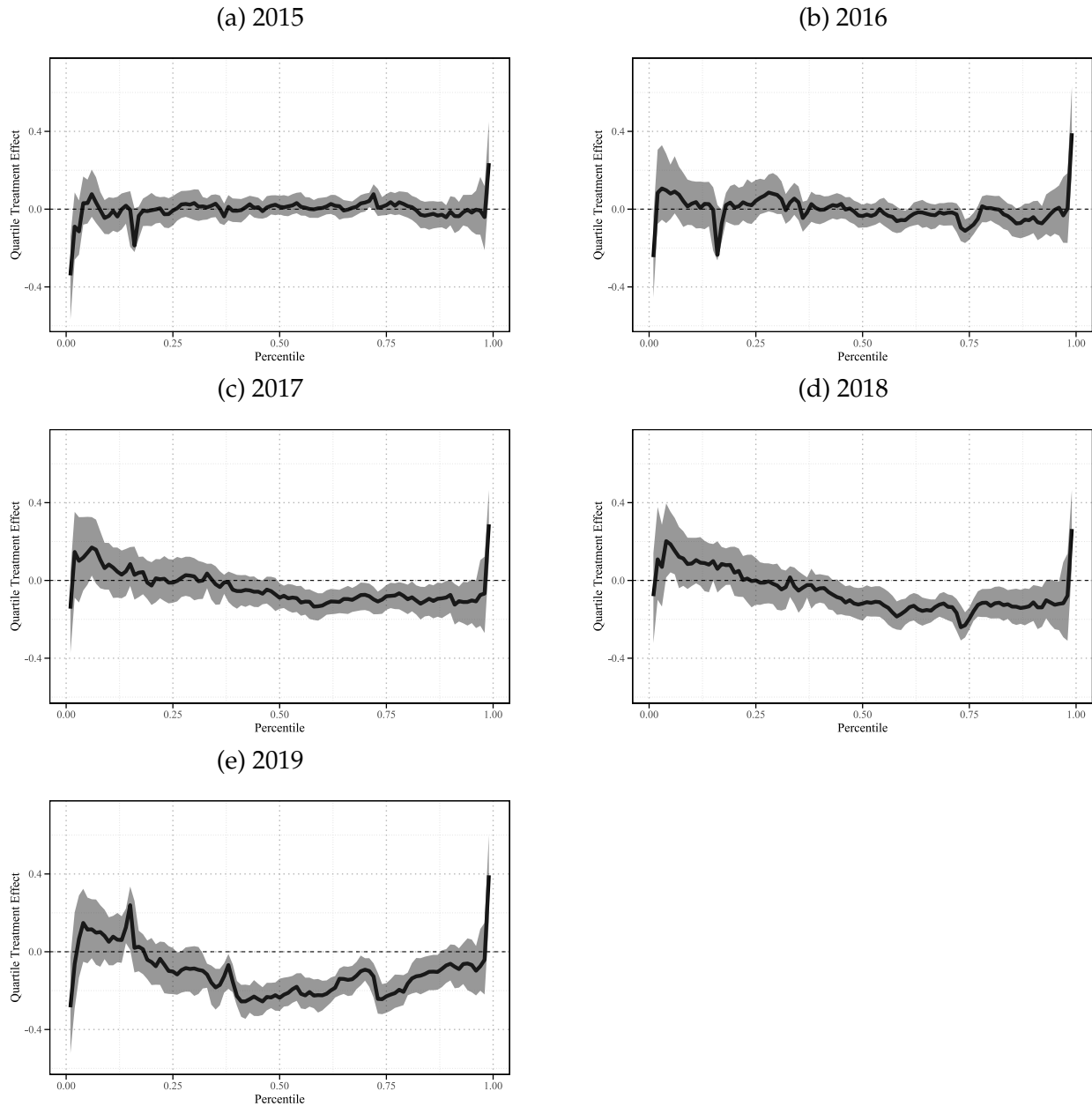


FIGURE B.7: Effect on Distribution of Capital-to-Intangible Ratio

This figure presents the results of a Change-in-Changes estimation using a sample of 115,529 firms with a total of 469,859 observations. The outcome variable is the logarithm of the ratio of physical capital at the end of the period (measured using PP&E) and intangible capital at the end of the period. For each percentile τ of the distribution of the outcome variable for treated firms in 2014 (the base year), we estimate the quantile treatment effect (QTE) between each year $t \geq 2015$ and the base year of 2014. We compute standard errors via a nonparametric block bootstrap that resamples firms (rather than individual observations) and present 95 percent confidence intervals.



C A Microfoundation for the Simple Model

In this section, we present a simple microfoundation for the wedge between the marginal rate of technical substitution and relative prices. A firm is characterized by its net worth $a > 0$ and productivity level z . It chooses the levels of physical capital k and intangible capital h to rent in order to produce output according to a Cobb-Douglas production function:

$$y = zk^\alpha h^\beta,$$

where $0 < \alpha + \beta < 1$, so the production technology exhibits decreasing returns to scale. The firm is a price taker in all markets, and the price of output is normalized to one.

The firm rents physical capital at price p_k and intangible capital at price p_h . While physical capital can be rented freely, access to intangible capital is subject to a financial constraint. Following [Moll \(2014\)](#), we assume that intangible input expenditures cannot exceed a fraction of net worth. That is, the firm must satisfy:

$$p_h h \leq \lambda a, \tag{C.1}$$

where $\lambda > 0$ captures the tightness of the financing constraint. A low value of λ corresponds to a more severe restriction on intangible capital use.

We solve the firm's problem in two steps. First, conditional on a given level of output—or scale— y , we solve for capital demand by minimizing costs. This cost minimization yields the optimal capital mix and the associated marginal cost of production. In the second step, we use marginal cost to determine the optimal scale y . The following Lemma characterizes the capital mix from the cost minimization step.

Lemma 1 *Let $\xi(y, a, z) \equiv \frac{k(y, a, z)}{h(y, a, z)}$ denote the optimal capital mix. Conditional on a given output level y , the cost minimization problem implies:*

$$\xi(y, a, z) = \begin{cases} \frac{\alpha}{\beta} \frac{p_h}{p_k} \cdot [1 + \mu(y, a, z)], & \text{if } a < \tilde{a}(y, z), \\ \frac{\alpha}{\beta} \frac{p_h}{p_k}, & \text{if } a \geq \tilde{a}(y, z), \end{cases} \tag{C.2}$$

where

$$1 + \mu(y, a, z) = \left(\frac{y}{z}\right)^{\frac{1}{\alpha}} \left(\frac{\lambda a}{p_h}\right)^{-\frac{\alpha+\beta}{\alpha}} \cdot \frac{\beta}{\alpha} \cdot \frac{p_k}{p_h}$$

is the Lagrange multiplier associated with the financial constraint. The constraint binds when

$a < \tilde{a}(y, z)$, where the threshold $\tilde{a}(y, z)$ is given by:

$$\tilde{a}(y, z) \equiv \left(\frac{y}{z}\right)^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha}{\beta} \cdot \frac{p_h}{p_k}\right)^{-\frac{\alpha}{\alpha+\beta}} \cdot \frac{p_h}{\lambda}. \quad (\text{C.3})$$

Proof. The firm solves

$$\min_{h,k} e^{-\tau} p_k k + e^{-\tau} p_h h \text{ s.to } zk^\alpha h^\beta \geq y, \quad e^{-\tau} p_h h \leq e^{\phi\tau} \lambda a$$

where the first order conditions are given by

$$\begin{aligned} e^{-\tau} p_k &= \theta \alpha \frac{y}{k} \\ e^{-\tau} p_h (1 + \mu) &= \theta \beta \frac{y}{h} \\ 0 &= \theta (y - zk^\alpha h^\beta) \\ 0 &= \mu (e^{-\tau} p_h h - e^{\phi\tau} \lambda a). \end{aligned}$$

The unconstrained firm has $\mu = 0$. Therefore, the optimal capital mix $\xi \equiv k/h$ is given by

$$\frac{p_k k}{p_h h} = \frac{\alpha}{\beta} \Leftrightarrow \xi = \frac{\alpha}{\beta} \frac{p_h}{p_k}.$$

Using the fact that $k = \xi h$ and the production function, we can write

$$y = z \xi^\alpha h^{\alpha+\beta} \Leftrightarrow h = \left(\frac{y}{z}\right)^{\frac{1}{\alpha+\beta}} \xi^{-\frac{\alpha}{\alpha+\beta}}.$$

As a consequence, the optimal level of capital is given by

$$k = \left(\frac{y}{z}\right)^{\frac{1}{\alpha+\beta}} \xi^{\frac{\beta}{\alpha+\beta}}.$$

Using these expressions, we can write total costs as

$$\begin{aligned} \mathcal{C} &= e^{-\tau} p_k k + e^{-\tau} p_h h \\ &= e^{-\tau} p_k \left(\frac{y}{z}\right)^{\frac{1}{\alpha+\beta}} \xi^{\frac{\beta}{\alpha+\beta}} + e^{-\tau} p_h \left(\frac{y}{z}\right)^{\frac{1}{\alpha+\beta}} \xi^{-\frac{\alpha}{\alpha+\beta}} \\ &= e^{-\tau} \left(\frac{y}{z}\right)^{\frac{1}{\alpha+\beta}} \left[p_k \xi^{\frac{\beta}{\alpha+\beta}} + p_h \xi^{-\frac{\alpha}{\alpha+\beta}} \right]. \end{aligned}$$

Under this expression, marginal costs are given by

$$\frac{dC}{dy} = \frac{1}{\alpha + \beta} \frac{1}{y} e^{-\tau} \left(\frac{y}{z} \right)^{\frac{1}{\alpha + \beta}} \left[p_k \xi^{\frac{\beta}{\alpha + \beta}} + p_h \xi^{-\frac{\alpha}{\alpha + \beta}} \right],$$

which are still increasing in y as we have decreasing returns to scale.

The constrained firm sets

$$h = e^{(1+\phi)\tau} \frac{\lambda a}{p_h}.$$

Using the production function, this implies that we can write physical capital as

$$y = z k^\alpha e^{\beta(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^\beta \Leftrightarrow k = \left(\frac{y}{z} \right)^{\frac{1}{\alpha}} e^{-\frac{\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{-\frac{\beta}{\alpha}}.$$

Therefore, the capital mix is given by

$$\xi = \frac{k}{h} = \left(\frac{y}{z} \right)^{\frac{1}{\alpha}} e^{-\frac{\alpha+\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{-\frac{\alpha+\beta}{\alpha}}.$$

Total costs are given by

$$C = e^{-\tau} p_k \left(\frac{y}{z} \right)^{\frac{1}{\alpha}} e^{-\frac{\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{-\frac{\beta}{\alpha}} + e^{-\tau} p_h e^{(1+\phi)\tau} \frac{\lambda a}{p_h}$$

where the marginal cost is given by

$$\frac{dC}{dy} = \frac{1}{\alpha} \frac{1}{y} e^{-\tau} p_k \left(\frac{y}{z} \right)^{\frac{1}{\alpha}} e^{-\frac{\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{-\frac{\beta}{\alpha}}$$

which is also increasing in y .

The firm is constrained if

$$\left(\frac{y}{z} \right)^{\frac{1}{\alpha + \beta}} \xi^{-\frac{\alpha}{\alpha + \beta}} > e^{(1+\phi)\tau} \frac{\lambda a}{p_h} \Leftrightarrow a < \left(\frac{y}{z} \right)^{\frac{1}{\alpha + \beta}} \xi^{-\frac{\alpha}{\alpha + \beta}} e^{-(1+\phi)\tau} \frac{p_h}{\lambda}$$

which defines our threshold.

We can therefore write the capital mix as

$$\xi(y, a, z, \tau) = \begin{cases} \left(\frac{y}{z}\right)^{\frac{1}{\alpha}} e^{-\frac{\alpha+\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h}\right)^{-\frac{\alpha+\beta}{\alpha}} & \text{if } a < \tilde{a}(y, z, \tau) \\ \frac{\alpha}{\beta} \frac{p_h}{p_k} & \text{if } a \geq \tilde{a}(y, z, \tau) \end{cases}$$

and the multiplier can be written as

$$\mu(y, a, z, \tau) = \frac{\xi(y, a, z, \tau)}{\frac{\alpha}{\beta} \frac{p_h}{p_k}}$$

which completes the proof. ■

The solution to the cost minimization problem implies that the capital mix $\xi(y, a, z)$ is decreasing in net worth: firms with higher net worth face looser financial constraints and converge to the efficient capital mix. The function $\xi(y, a, z)$ is continuous but non-differentiable at the threshold $\tilde{a}(y, z)$, where a kink emerges due to the transition from constrained to unconstrained status.

Throughout the paper, we refer to firms with $a < \tilde{a}(y, z)$ as *constrained*, and firms with $a \geq \tilde{a}(y, z)$ as *unconstrained*. Unconstrained firms all implement the frictionless capital ratio $\frac{\alpha}{\beta} \cdot \frac{p_h}{p_k}$.

The multiplier $\mu(y, a, z)$ associated with the financial constraint is decreasing in both net worth a and the pledgability of intangible capital λ , reflecting the idea that greater internal resources or collateral capacity relax the constraint. However, $\mu(y, a, z)$ is increasing in scale y : holding productivity fixed, larger scale raises the demand for intangibles and tightens the collateral constraint. A similar logic applies to the threshold $\tilde{a}(y, z)$, which increases with scale y conditional on productivity z , implying that some firms may become constrained as they expand.

The cost minimization problem also yields a cost function $\mathcal{C}(y, a, z)$ that is continuous but exhibits a kink at $a = \tilde{a}(y, z)$ due to the change in constraint status. Since the firm is a price taker in the output market and we normalize the price of output to one, the optimal scale y^* is pinned down by the first-order condition $C_y(y^*, a, z) = 1$. The following lemma characterizes the optimal scale.

Lemma 2 *The optimal scale $y^*(a, z)$ is given by:*

$$y^*(a, z) = \begin{cases} \alpha^{\frac{\alpha}{1-\alpha}} z^{\frac{1}{1-\alpha}} p_k^{-\frac{1}{1-\alpha}} \left(\frac{\lambda a}{p_h} \right)^{\frac{\beta}{1-\alpha}}, & \text{if } a < \tilde{a}(y^*(a, z), z), \\ (\alpha + \beta)^{\frac{\alpha+\beta}{1-\alpha-\beta}} z^{\frac{1}{1-\alpha-\beta}} \left[p_k \xi_u^{\frac{\beta}{\alpha+\beta}} + p_h \xi_u^{-\frac{\alpha}{\alpha+\beta}} \right]^{-\frac{\alpha+\beta}{1-\alpha-\beta}}, & \text{if } a \geq \tilde{a}(y^*(a, z), z), \end{cases} \quad (\text{C.4})$$

where $\xi_u \equiv \frac{\alpha}{\beta} \cdot \frac{p_h}{p_k}$ denotes the unconstrained capital mix.

Proof. The profit maximization problem is given by

$$\max_y y - \mathcal{C}(y)$$

where $\mathcal{C}(y)$ is the cost function.

Suppose the firm is unconstrained. Then, it follows that $\mathcal{C}'(y) = 1$ which implies

$$\begin{aligned} &\Leftrightarrow \frac{1}{\alpha + \beta} \frac{1}{y} e^{-\tau} \left(\frac{y}{z} \right)^{\frac{1}{\alpha+\beta}} \left[p_k \xi^{\frac{\beta}{\alpha+\beta}} + p_h \xi^{-\frac{\alpha}{\alpha+\beta}} \right] = 1 \\ &\Leftrightarrow \frac{1}{y} \left(\frac{y}{z} \right)^{\frac{1}{\alpha+\beta}} = (\alpha + \beta) e^{\tau} \left[p_k \xi^{\frac{\beta}{\alpha+\beta}} + p_h \xi^{-\frac{\alpha}{\alpha+\beta}} \right]^{-1} \\ &\Leftrightarrow y^{\frac{1-\alpha-\beta}{\alpha+\beta}} = z^{\frac{1}{\alpha+\beta}} (\alpha + \beta) e^{\tau} \left[p_k \xi^{\frac{\beta}{\alpha+\beta}} + p_h \xi^{-\frac{\alpha}{\alpha+\beta}} \right]^{-1} \\ &\Leftrightarrow y = z^{\frac{1}{1-\alpha-\beta}} (\alpha + \beta)^{\frac{\alpha+\beta}{1-\alpha-\beta}} e^{\frac{\alpha+\beta}{1-\alpha-\beta} \tau} \left[p_k \xi^{\frac{\beta}{\alpha+\beta}} + p_h \xi^{-\frac{\alpha}{\alpha+\beta}} \right]^{-\frac{\alpha+\beta}{1-\alpha-\beta}}. \end{aligned}$$

Instead, if the firm is constrained, it follow that

$$\begin{aligned} &\Leftrightarrow \frac{1}{\alpha} \frac{1}{y} e^{-\tau} p_k \left(\frac{y}{z} \right)^{\frac{1}{\alpha}} e^{-\frac{\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{-\frac{\beta}{\alpha}} = 1 \\ &\Leftrightarrow \frac{1}{y} \left(\frac{y}{z} \right)^{\frac{1}{\alpha}} = e^{\frac{\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{\frac{\beta}{\alpha}} \alpha e^{\tau} \frac{1}{p_k} \\ &\Leftrightarrow y^{\frac{1-\alpha}{\alpha}} = e^{\frac{\beta}{\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{\frac{\beta}{\alpha}} \alpha e^{\tau} \frac{1}{p_k} z^{\frac{1}{\alpha}} \\ &\Leftrightarrow y = e^{\frac{\beta}{1-\alpha}(1+\phi)\tau} \left(\frac{\lambda a}{p_h} \right)^{\frac{\beta}{1-\alpha}} \alpha^{\frac{\alpha}{1-\alpha}} e^{\frac{\alpha}{1-\alpha}\tau} p_k^{-\frac{\alpha}{1-\alpha}} z^{\frac{1}{1-\alpha}} \end{aligned}$$

which completes the proof. ■

Consider first the constrained firms. Their optimal scale increases with net worth and

with the pledgability of intangible capital λ . In contrast, for unconstrained firms, optimal scale is independent of both net worth and λ ; it depends only on productivity z and input prices. This reflects that, once the constraint is slack, scale is determined solely by technology and prices.

D Constructing the Distribution of the Wedge

Let $X \sim F$ and let $U \equiv F(X)$ denote the percentile rank of X . By the probability integral transform, $U \sim \text{Unif}(0,1)$ under continuity of F ; if F has atoms, a randomized rank $U = F(X^-) + V[F(X) - F(X^-)]$ with $V \sim \text{Unif}(0,1)$ ensures the same result. We are interested in the distribution of

$$Y = g(U),$$

where $g : [0,1] \rightarrow \mathbb{R}$ denotes the wedge function estimated or tabulated across percentiles.

D.1 Piecewise-linear representation of g

The function g is observed on a grid of percentiles $0 \leq p_1 < \dots < p_N \leq 1$ with corresponding values $g_i = g(p_i)$. We sort by p , drop duplicates, and pad the endpoints so that $p_1 = 0$ and $p_N = 1$ if needed. Between consecutive grid points, we linearly interpolate:

$$g(p) = g_i + \frac{g_{i+1} - g_i}{p_{i+1} - p_i} (p - p_i).$$

For each segment, define the width $\Delta p_i = p_{i+1} - p_i$ and slope $\Delta g_i = g_{i+1} - g_i$.

D.2 Distribution induced by linear segments

Conditioning on the segment $[p_i, p_{i+1}]$, the probability that U falls in this interval is Δp_i . If $\Delta g_i \neq 0$, the linear map from p to y is invertible, and Y is uniform on the image interval

$$[y_{\min,i}, y_{\max,i}] = [\min(g_i, g_{i+1}), \max(g_i, g_{i+1})].$$

The conditional density on this interval is constant and equal to

$$f_{Y|i}(y) = \frac{\Delta p_i}{|\Delta g_i|}.$$

If $\Delta g_i = 0$, the entire probability mass Δp_i collapses to the point $y = g_i$, creating an atom.

Aggregating across segments yields the piecewise-constant density for the continuous part:

$$f_Y(y) = \sum_{i: \Delta g_i \neq 0} \frac{\Delta p_i}{|\Delta g_i|} \mathbf{1}\{y \in [y_{\min,i}, y_{\max,i}]\},$$

and point masses at the levels where g is flat:

$$\Pr(Y = c) = \sum_{i: \Delta g_i = 0, g_i = c} \Delta p_i.$$

The corresponding CDF is

$$F_Y(y) = \sum_{i: \Delta g_i \neq 0} \Delta p_i \frac{(y - y_{\min, i})_+}{y_{\max, i} - y_{\min, i}} \wedge \Delta p_i + \sum_{i: \Delta g_i = 0} \Delta p_i \mathbf{1}\{y \geq g_i\},$$

where $a_+ = \max(a, 0)$ and $x \wedge b = \min(x, b)$.